ETAS RTA-BSW v12.7.0

RTA-Safe Stack Reference Guide

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1 End to End Library

Module name	E2E
Package	RTA-SAFE
AUTOSAR Module ID	207
AUTOSAR Version	22-11

1.1 Introduction

The End to End Library (E2E) is in the system services abstraction layer, forming part of the safety (RTA-SAFE) module stack. The diagram below shows the position of E2E in the AUTOSAR layered architecture.

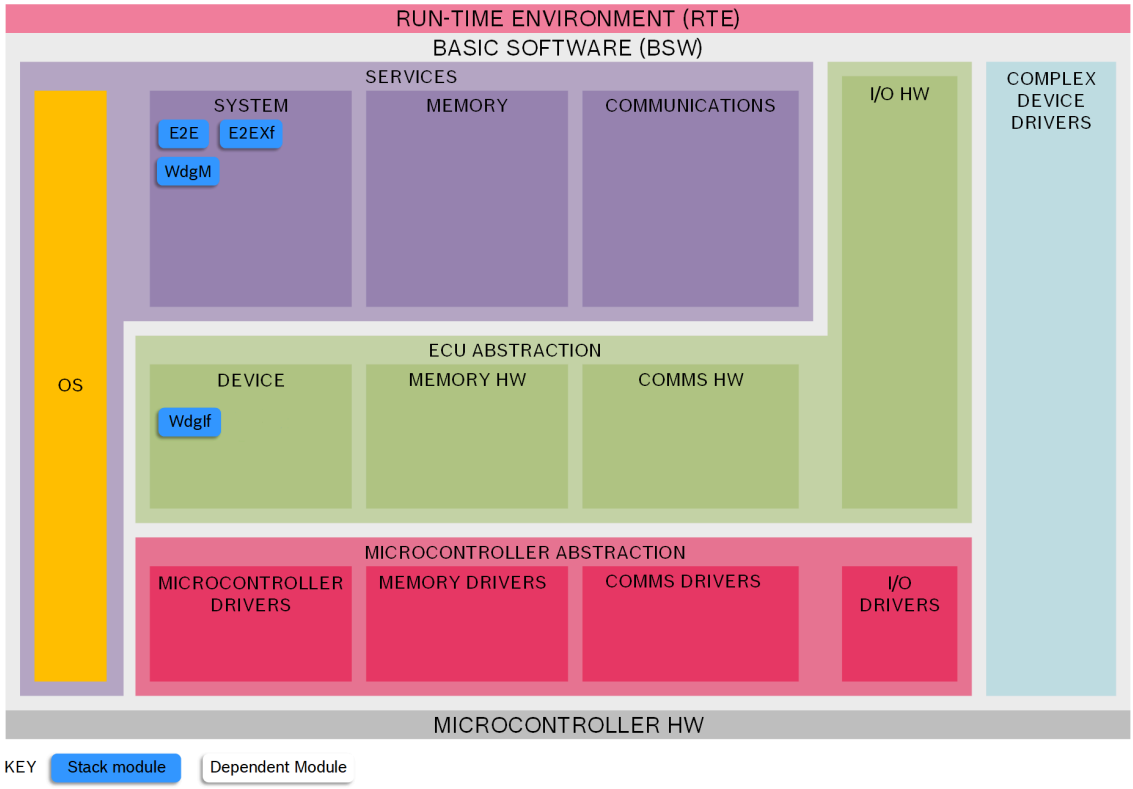

Figure 1.1.RTA-SAFE Architecture

The purpose of the E2E component is to provide protection mechanisms for end-to-end communication. The E2E library offers software routines that safeguard the data exchange between software components, applicable to both client-server and sender-receiver communication, whether inter-ECU or intra-ECU. The library includes mechanisms for failure detection and authentication.

It is designed to support various communication levels, including both data element level and I-PDU level. Additionally, it is not limited to a specific communication channel, allowing for communication through CAN, FlexRay, LIN, or

intra-ECU pipelines.

Therefore, the E2E library implements protection mechanisms on the client or sender side and verification mechanisms on the server or receiver side.

- On the sender side, data protection takes place on the raw data stream, which is prepared for secure communication by the E2EProtect routines.
- On the receiver side, after transmission, the data stream is analysed with respect to correctness of the content, correctness of the sender-receiver relation, and correctness of the sequence of the message. This analysis is performed by an E2ECheck routine in combination with a CRC routine (see below).

1.1.1 Crc Library

Most of the end to end communication protection is based on the utilisation of cyclic redundancy checks, which are provided from a Crc library. See the Supported Features section for more details.

1.1.2 E2E Profiles

The E2E library provides E2E profiles for end-to-end communication protection. These profiles are groups of routines that provide mechanisms for the recognition of data corruption, addressing faults, incorrect sequence of messages and timeouts. These profiles provide three main groups of routines:

- The E2EProtect routines, called from the sender.
- The E2ECheck routines, called from the receiver.
- Status mapping routines used for generic status reporting on the receiver side.

Module Configuration

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

1.2 Supported Features

1.2.1 E2E Profiles

The RTA-BSW implementation of E2E is based on the AUTOSAR 4.2 specification, but with anticipated functions from AUTOSAR 4.3 (e.g. support for Profiles 04, 07, 11, 22 and 44). As such, E2E supports the following profiles to provide data protection and check functions:

1.2.1.1 E2E Profile 1

Works with the 8-bit SAE J1850 routine from the Crc library. It is intended to protect and check data streams of a maximum length of 30 bytes when the number of messages in a system is small. It also checks the authenticity of the sender and the sequence of the messages.

1.2.1.2 E2E Profile 2

Works with the 8-bit Crc8h2f from the Crc library. It is intended to protect and check data streams of a maximum length of 256 bytes when the number of messages in a system is high. This profile avoids masquerading of messages. The sequence error will be detected with a delay.

1.2.1.3 E2E Profile 4

Works with the 32-bit Crc32P4 from the Crc library. It is intended to protect and check data streams of a maximum length of 4096 bytes. Small number of messages for big data streams.

1.2.1.4 E2E Profile 5

Works with the 16-bit Crc16 from the Crc library. It is intended to protect and check data streams of a maximum length of 4096 bytes. The number of messages can be high. This profile avoids masquerading of messages.

1.2.1.5 E2E Profile 6

Works with the 16-bit Crc16 from the Crc library. It is intended to protect and check data streams with variable data length but with a maximum length of 4096 bytes. The number of messages can be high. This profile avoids masquerading of messages.

1.2.1.6 E2E Profile 7

Works with the 64-bit CRC64 from the Crc library. It is optimized for communication protection for communication via Ethernet, and it is designed for systems with high protection requirements and a low number of messages.

1.2.1.7 E2E Profile 8

Is optimized for communication protection in case of communication via Ethernet. It is designed for systems with high protection requirements and low message numbers. It works with the 32-bit CRC32P4 from the Crc library and supports a variable length of data.

1.2.1.8 E2E Profile 11

Works with the 8-bit SAE J1850 routine from the Crc library. It is optimized for communication protection for communication via CAN, FlexRay and Lin, and it is designed for systems with high protection requirements and a low number of messages.

1.2.1.9 E2E Profile 22

Works with the 8-bit Crc8h2f from the Crc library. It is optimized for communication protection for communication via CAN, FlexRay and Lin, and it is designed for systems with low protection requirements and a high number of messages.

1.2.1.10 E2E Profile 44

Works with the 32-bit CRC32P4 from the CRC library. It is optimized for communication protection in case of communication via Ethernet. It is designed for systems with high protection requirements and low numbers of messages.

1.2.1.11 E2E Profile 4m

Is optimized for communication protection in case of communication via Ethernet. It is designed for systems with high protection requirements and low message numbers. It works with the CRC32P4 calculation from the CRC library and supports a variable length of data.

1.2.1.12 E2E Profile 7m

Is optimized for communication protection in case of communication via Ethernet. It is designed for systems with high protection requirements and low message numbers. It works with the 64-bit CRC64 from the CRC library and supports a variable length of data.

1.2.2 E2E State Machine

The E2E State Machine works with the available E2E profiles and it extends the profile-provided check functionality.

1.2.3 Utilisation of Crc

The E2EProtect routines use complex cyclic redundancy checks over the data stream and the data ID. The data ID is an identifier for the sender and it is responsible for the sender-receiver authentication.

The result of the CRC calculation is added to the sender data stream before transmission. The E2ECheck routine, performed by the receiver, calculates a complex CRC over the received data stream and the data ID, and it then compares the result with the transmitted CRC result. If the CRC results are identical, a correct message and a correct sender-receiver authentication is assumed.

E2E thus requires an additional library for the Crc calculations. It can use either the Crc library provided as part of the RTA-BSW E2Eservices stack, which is automatically called from the E2E functions, or it can use an external Crc Library (e.g. if non-Autosar Crc functionality is required).

E2E recognizes the available Crc library and automatically corrects a difference in the requested and provided START and XOR values with an additional XOR command. If an automatic correction is not possible a manual correction can be done instead.

The Crc library, and the details for the manual correction, can be configured (see the Configuration Section for more details).

The following interfaces of Crc are used by E2E:

- Crc_CalculateCRC8 (used by Profile 1 and Profile 11)
- Crc_CalculateCRC8H2F (used by Profile 2 and Profile 22)

- Crc_CalculateCRC32P4 (used by Profile 4, Profile 44, Profile 8 and Profile 4m)
- Crc_CalculateCRC16 (used by Profile 5 and Profile 6)
- Crc_CalculateCRC64 (used by Profile 7 and Profile 7m)

1.3 Extensions

1.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

1.3.1.1 E2E

Short Name	E2E
Description	RB specific configuration of the E2E (E2E routines) module.
Multiplicity	0..1
Default Value	-
Type	ECUC-MODULE-DEF

1.3.1.2 E2E/E2ERbGMProfiles

Short Name	E2ERbGMProfiles
Long Name	E2E Custom GM Profiles
Description	Configuration container for E2E Custom GM profiles
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.3 E2E/E2ERbGMProfiles/E2ERbGM_Profile01

Short Name	E2ERbGM_Profile01
Description	Enables / disables GM E2E Profile 01
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.4 E2E/E2ERbGMProfiles/E2ERbGM_Profile02

Short Name	E2ERbGM_Profile02
Description	Enables / disables GM E2E Profile 02
Multiplicity	0..1

Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.5 E2E/E2ERbGMProfiles/E2ERbGM_Profile03

Short Name	E2ERbGM_Profile03
Description	Enables / disables GM E2E Profile 03
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.6 E2E/E2ERbGMProfiles/E2ERbGM_Profile04

Short Name	E2ERbGM_Profile04
Description	Enables / disables GM E2E Profile 04
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.7 E2E/E2ERbGeneral

Short Name	E2ERbGeneral
Long Name	E2E general configurations
Description	General configuration of component E2E. The general configuration defines the functions, provided by the E2E module.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.8 E2E/E2ERbGeneral/E2ERbCallout

Short Name	E2ERbCallout
Long Name	Callout impl.
Description	Enables callout feature in E2E
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.9 E2E/E2ERbGeneral/E2ERbCheckParameterType

Short Name	E2ERbCheckParameterType
Long Name	Check Parameter Type (E2E_P0XConfigType)
Description	Defines the method to check the contents of the configuration structure of type E2E_P0XConfigType-.E2E_INTEGRATED: Check is integrated within the E2E_P0XProtect/E2E_P0XCheck functions. This means the check is executed every time E2E_P0XProtect or E2E_P0XCheck is called and will increase the security by preventing unexpected behavior of the function in case of a misconfiguration but slightly increase the runtime.E2E_MANUAL: Check is available as additional function. The function rba_E2E_P0XCheckConfig-Type() can be called once or multiple times (depends on the use case) before E2E_P0XProtect/E2E_P0XCheck is called.E2E_NONE: Additional checks are disabled. This will slighty reduce the runtime of E2E_P0XProtect/E2E_P0XCheck.
Multiplicity	1..1
Default Value	E2E_INTEGRATED
Enumeration Literals	E2E_INTEGRATED, E2E_MANUAL, E2E_NONE
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.10 E2E/E2ERbGeneral/E2ERbCheckStatusTypeCompatibilityMode

Short Name	E2ERbCheckStatusTypeCompatibilityMode
Long Name	Check Status Type Compatibility Mode (E2E_P0XCheckStatusType)
Description	Switch to enable/disable the compatibility to AUTOSAR for E2E_P01CheckStatusType and E2E_P02CheckStatusType. If compatibility mode is enabled then the E2E_P0XCheckStatusType is implemented like originaly in the AUTOSAR specification defined (default). If the compatibility mode is disabled the E2E_P0XCheckStatusType is implemented like originaly in the Autosar specification before 4.2 defined to use similar values like in Profile 5 and 6.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.11 E2E/E2ERbGeneral/E2ERbCrcManualDefinitions

Short Name	E2ERbCrcManualDefinitions
Long Name	Crc manual definitions
Description	This container contains the START- and XOR-values for the manual correction of the crc calculation.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.12 E2E/E2ERbGeneral/E2ERbCrcManualDefinitions/E2ERbCrc8FirstCall

Short Name	E2ERbCrc8FirstCall
Long Name	Crc8 firstCall configuration
Description	It can be configured if the Crc library provides the parameter Crc_IsFirstCall in the interface for the crc8 calculation.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.13 E2E/E2ERbGeneral/E2ERbCrcManualDefinitions/E2ERbCrc8ManualValues

Short Name	E2ERbCrc8ManualValues
Long Name	Crc8 START-/XOR-values
Description	If manual correction for the crc calculation is enabled, the Crc8 START- and XOR-values must be given here.
Multiplicity	1..1
Default Value	xFF
Enumeration Literals	xFF, x00
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.14 E2E/E2ERbGeneral/E2ERbDataIDBitOrderProfile5

Short Name	E2ERbDataIDBitOrderProfile5
Long Name	E2E profile 5: bit order of Data ID
Description	Defines if the bit order of the Data ID used for CRC calculation is changed.ORDER_UNCHANGED: order is not changedORDER_FLIPPED: order is flipped

Multiplicity	1..1
Default Value	ORDER_UNCHANGED
Enumeration Literals	ORDER_UNCHANGED, ORDER_FLIPPED
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.15 E2E/E2ERbGeneral/E2ERbDataIDBitOrderProfile6

Short Name	E2ERbDataIDBitOrderProfile6
Long Name	E2E profile 6: bit order of Data ID
Description	Defines if the bit order of the Data ID used for CRC calculation is changed.ORDER_UNCHANGED: order is not changedORDER_FLIPPED: order is flipped
Multiplicity	1..1
Default Value	ORDER_FLIPPED
Enumeration Literals	ORDER_UNCHANGED, ORDER_FLIPPED
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.16 E2E/E2ERbGeneral/E2ERbDevTest

Short Name	E2ERbDevTest
Long Name	Additional testing defines for unit tests
Description	Generate additional defines for testing. This parameter needs to be disabled in production code! Default: false
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.17 E2E/E2ERbGeneral/E2ERbHeaderBitOrderProfile11

Short Name	E2ERbHeaderBitOrderProfile11
Long Name	E2E profile 11: bit order of header information
Description	Defines if the bit order of elements within the message header is changed. ORDER_UNCHANGED: order is not changed. ORDER_FLIPPED: order is flipped
Multiplicity	1..1
Default Value	ORDER_FLIPPED
Enumeration Literals	ORDER_UNCHANGED, ORDER_FLIPPED
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.18 E2E/E2ERbGeneral/E2ERbHeaderBitOrderProfile22

Short Name	E2ERbHeaderBitOrderProfile22
Long Name	E2E profile 22: bit order of header information
Description	Defines if the bit order of elements within the message header is changed.ORDER_UNCHANGED: order is not changedORDER_FLIPPED: order is flipped
Multiplicity	1..1
Default Value	ORDER_FLIPPED
Enumeration Literals	ORDER_UNCHANGED, ORDER_FLIPPED
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.19 E2E/E2ERbGeneral/E2ERbHeaderBitOrderProfile4

Short Name	E2ERbHeaderBitOrderProfile4
Long Name	E2E profile 4: bit order of header information
Description	Defines if the bit order of elements within the message header is changed.ORDER_UNCHANGED: order is not changedORDER_FLIPPED: order is flipped
Multiplicity	1..1
Default Value	ORDER_FLIPPED
Enumeration Literals	ORDER_UNCHANGED, ORDER_FLIPPED
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.20 E2E/E2ERbGeneral/E2ERbHeaderBitOrderProfile44

Short Name	E2ERbHeaderBitOrderProfile44
Long Name	E2E profile 44: bit order of header information
Description	Defines if the bit order of elements within the message header is changed.ORDER_UNCHANGED: order is not changedORDER_FLIPPED: order is flipped
Multiplicity	1..1
Default Value	ORDER_FLIPPED
Enumeration Literals	ORDER_UNCHANGED, ORDER_FLIPPED
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.21 E2E/E2ERbGeneral/E2ERbHeaderBitOrderProfile5

Short Name	E2ERbHeaderBitOrderProfile5
Long Name	E2E profile 5: bit order of header information
Description	Defines if the bit order of elements within the message header is changed.ORDER_UNCHANGED: order is not changedORDER_FLIPPED: order is flipped
Multiplicity	1..1
Default Value	ORDER_UNCHANGED
Enumeration Literals	ORDER_UNCHANGED, ORDER_FLIPPED
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.22 E2E/E2ERbGeneral/E2ERbHeaderBitOrderProfile6

Short Name	E2ERbHeaderBitOrderProfile6
Long Name	E2E profile 6: bit order of header information
Description	Defines if the bit order of elements within the message header is changed.ORDER_UNCHANGED: order is not changedORDER_FLIPPED: order is flipped
Multiplicity	1..1
Default Value	ORDER_FLIPPED
Enumeration Literals	ORDER_UNCHANGED, ORDER_FLIPPED
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.23 E2E/E2ERbGeneral/E2ERbHeaderBitOrderProfile7

Short Name	E2ERbHeaderBitOrderProfile7
Long Name	E2E profile 7: bit order of header information
Description	Defines if the bit order of elements within the message header is changed.ORDER_UNCHANGED: order is not changedORDER_FLIPPED: order is flipped
Multiplicity	1..1
Default Value	ORDER_FLIPPED
Enumeration Literals	ORDER_UNCHANGED, ORDER_FLIPPED
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.24 E2E/E2ERbGeneral/E2ERbHeaderBitOrderProfile8

Short Name	E2ERbHeaderBitOrderProfile8
Long Name	E2E profile 8: bit order of header information
Description	Defines if the bit order of elements within the message header is changed. ORDER_UNCHANGED: order is not changed, ORDER_FLIPPED: order is flipped
Multiplicity	1..1
Default Value	ORDER_FLIPPED
Enumeration Literals	ORDER_UNCHANGED, ORDER_FLIPPED
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.25 E2E/E2ERbGeneral/E2ERbProfile1

Short Name	E2ERbProfile1
Long Name	E2E profile 1 configuration
Description	Switch to enable/disable the availability of the E2E profile 1 communication protection method. If E2ERbProfile1 calculation is enabled the Crc calculation method Crc8 must be selected in the configuration of the module Crc. For E2ERbProfile1 the default value is “enabled”.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.26 E2E/E2ERbGeneral/E2ERbProfile11

Short Name	E2ERbProfile11
Long Name	E2E profile 11 configuration
Description	Switch to enable/disable the availability of the E2E profile 11 communication protection method. If E2ERbProfile11 calculation is enabled the Crc calculation method Crc8 must be selected in the configuration of the module Crc. For E2ERbProfile11 the default value is “enabled”.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.27 E2E/E2ERbGeneral/E2ERbProfile2

Short Name	E2ERbProfile2
Long Name	E2E profile 2 configuration
Description	Switch to enable/disable availability of the E2E profile 2 communication protection method. If E2ERbProfile2 calculation is enabled the Crc calculation method Crc8 must be selected in the configuration of the module Crc. For E2ERbProfile2 the default value is “enabled”.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.28 E2E/E2ERbGeneral/E2ERbProfile22

Short Name	E2ERbProfile22
Long Name	E2E profile 22 configuration
Description	Switch to enable/disable availability of the E2E profile 22 communication protection method. If E2ERbPro-file22 calculation is enabled the Crc calculation method Crc8 must be selected in the configuration of the module Crc. For E2ERbProfile22 the default value is “enabled”.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.29 E2E/E2ERbGeneral/E2ERbProfile4

Short Name	E2ERbProfile4
Long Name	E2E profile 4 configuration
Description	Switch to enable/disable availability of the E2E profile 4 communication protection method. If E2ERbProfile4 calculation is enabled the Crc calculation method Crc32P4 must be selected in the configuration of the module Crc. For E2ERbProfile4 the default value is “enabled”.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.30 E2E/E2ERbGeneral/E2ERbProfile44

Short Name	E2ERbProfile44
Long Name	E2E profile 44 configuration
Description	Switch to enable/disable availability of the E2E profile 44 communication protection method. If E2ERbProfile44 calculation is enabled the Crc calculation method Crc32P4 must be selected in the configuration of the module Crc. For E2ERbProfile44 the default value is “enabled”.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.31 E2E/E2ERbGeneral/E2ERbProfile4m

Short Name	E2ERbProfile4m
Long Name	E2E profile 4m configuration
Description	Switch to enable/disable availability of the E2E profile 4m communication protection method. If E2ERbProfile4m calculation is enabled the Crc calculation method Crc32P4 must be selected in the configuration of the module Crc. For E2ERbProfile4m the default value is “enabled”.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.32 E2E/E2ERbGeneral/E2ERbProfile5

Short Name	E2ERbProfile5
Long Name	E2E profile 5 configuration
Description	Switch to enable/disable availability of the E2E profile 5 communication protection method. If E2ERbProfile5 calculation is enabled the Crc calculation method Crc16 must be selected in the configuration of the module Crc. For E2ERbProfile5 the default value is “enabled”.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.33 E2E/E2ERbGeneral/E2ERbProfile6

Short Name	E2ERbProfile6
Long Name	E2E profile 6 configuration
Description	Switch to enable/disable availability of the E2E profile 6 communication protection method. If E2ERbProfile6 calculation is enabled the Crc calculation method Crc16 must be selected in the configuration of the module Crc. For E2ERbProfile6 the default value is “enabled”.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.34 E2E/E2ERbGeneral/E2ERbProfile7

Short Name	E2ERbProfile7
Long Name	E2E profile 7 configuration
Description	Switch to enable/disable availability of the E2E profile 7 communication protection method. If E2ERbProfile7 calculation is enabled the Crc calculation method Crc64 must be selected in the configuration of the module Crc. For E2ERbProfile7 the default value is “enabled”.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.35 E2E/E2ERbGeneral/E2ERbProfile7m

Short Name	E2ERbProfile7m
Long Name	E2E profile 7m configuration
Description	Switch to enable/disable availability of the E2E profile 7m communication protection method. If E2ERbPro-file7m calculation is enabled the Crc calculation method Crc64 must be selected in the configuration of the module Crc. For E2ERbProfile7m the default value is “disabled”.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.36 E2E/E2ERbGeneral/E2ERbProfile8

Short Name	E2ERbProfile8
Long Name	E2E profile 8 configuration
Description	Switch to enable/disable availability of the E2E profile 8 communication protection method. If E2ERbProfile8 calculation is enabled the Crc calculation method Crc32P4 must be selected in the configuration of the module Crc. For E2ERbProfile8 the default value is “enabled”.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.37 E2E/E2ERbGeneral/E2ERbSMVersion

Short Name	E2ERbSMVersion
Long Name	Choose the AUTOSAR version for E2E state machine
Description	This parameter is provided to select the state machine behaviour. This is an enumertaion parameter. It contains two values : AR42 and AR45. If selected AR42, the SM behaves according to AR42 E2E SWS. If selected AR45, the SM behaves as per R19-11 or later versions. Default value : AR42
Multiplicity	0..1
Default Value	AR42
Enumeration Literals	AR42, AR45
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.38 E2E/E2ERbGeneral/E2ERbStateMachine

Short Name	E2ERbStateMachine
Long Name	E2E state machine configuration
Description	Switch to enable/disable availability of the generic state handling methods.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.39 E2E/E2ERbGeneral/E2ERbVersionInfoApi

Short Name	E2ERbVersionInfoApi

Long Name	E2E version information API configuration
Description	Enables the functions for reading the module version information. For E2EVersionInfoApi the default value is “enabled”. For disabling of E2EVersionInfoApi voting is required.
Multiplicity	1..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.40 E2E/E2ERbGeneral/E2ERbXORProfile11

Short Name	E2ERbXORProfile11
Long Name	Enable additional XOR opreation for E2E profile 11
Description	Switch to enable/disable the availability of the additional XOR operation for Protect and Check Functions as per Autosar version R22-11 onwards.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

- There are no additional API defined in this version.

1.4 Limitations

The E2E module has the following deviations from AR v22-11:

Configuration Mechanism

In the ECU Navigator, the following configurations can be done:

- Provision of E2E version information (On-Off).
- Selection of provided E2E profiles.
- Selection of header bit order for profiles 04, 05, 06, 07, 08, 11, 22 and 44 (ORDER_UNCHANGED or ORDER_FLIPPED).
- Selection of Data Id bit order for profiles 05 and 06 (ORDER_UNCHANGED or ORDER_FLIPPED).
- Configuration of Crc recognition mechanism.
- Generic state machine (On-Off).
- Compatibility mode for `E2E_P01CheckStatusType` and `E2E_P02CheckStatusType` (On-Off).
- Configuration check handling method selection (INTERNAL, EXTERNAL,

NONE).

- Configuration of test defines used for unit tests (On-Off (must be disabled for production code)).

MemMap

In contrast to the AR specification for the E2E library, a component-specific MemMap header file *E2E_Memmap.h* is required. This header file is included directly before, or at the end of, C functions or constants. Therefore, each function or constant can be located separately and it is possible to specify different memory areas. This deviates from the AR E2E specification, but it is consistent with the AR specification for memory mapping. In other words, the AR specifications for the E2E library and memory mapping are conflicting.

E2E_Prv_<file>

In contrast to the AUTOSAR specification of the E2E library, inline functions are outsourced to other private files. This is to reduce the complexity and to encapsulate the private C objects.

Call of Crc functions

In contrast to the AUTOSAR specification, which required identical revisions for the E2E and Crc libraries, this E2E implementation works with each AR conformal Crc library, which provides the requested Crc interfaces.

CheckInit value changes for profiles 1 and 2

This deviation modifies the initial value of `WaitForFirstData` in `E2E_P<01|02>CheckInit` (TRUE instead of FALSE) to get a robust behavior for the first message. The status is mapped to OK or WRONGSEQUENCE only, instead of OK, OKSOMELOST, REPEATED or WRONGSEQUENCE.

Contradictory initialization of counter value in profiles 05 and 06

The counter in `E2E_P<05|06>CheckStateType` could be initialized with 0 or 0xFF (see SWS_E2E_00451/459 and SWS_E2E_00472/473). The decision was made that the counter is initialized with 0xFF because the status is E2E_P05STATUS_OK, instead of E2E_P05STATUS_WRONGSEQUENCE, if the sender and receiver are already synchronized with the first message, which appears more meaningful.

Profile 02 counter increment position

Moves the message counter increment in `E2E_P02Protect` to the end of the function so that the behavior of Profile 02 is more coherent with other profiles.

1.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- There are no known exclusions in this version.

1.6 Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

- There are no known deviations in this version.

1.7 API Definition

The E2E Library does not contain any APIs.

1.8 Configuration Advice

NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

1.8.1 Manual Recognition of Crc

Additional configuration is necessary if automatic recognition of the Crc software needs to be replaced by a manual recognition.

To enable external Crc library usage mode, the E2ERbCrcManualDefinitions sub container needs to be created.

- The E2ERbCrc8FirstCall parameter must be set to TRUE if the Crc8 software provides an additional parameter to select the first call of the function.
- The E2ERbCrc8ManualValues parameter is where the Crc8 START and XOR values provided by the Crc software are selected.

1.8.2 E2E Custom Profiles

Profile 01 and Profile 02 are implemented as prototype features. To integrate the E2E Custom Profiles feature:

1. Instead of E2E_EcucValues.arxml, register E2E_EcucValues_GM.arxml
2. Instead of Crc_EcucValues.arxml, register Crc_EcucValues_GM.arxml for E2E GM Profile03.

- The E2E Custom Profile feature is supported with a new configuration container(E2ERbGMProfiles) under E2E parent container. The E2E Custom profile generation is only created if the required GM Profiles are enabled in this container.
- The E2E Custom Profile 03 uses custom CRC8 Calculation. CrcGM container under Crc module needs to be enabled to support CRC calculation in Profile

03.

1.9 Integration Advice

1.9.1 Init Functions

No specific handling necessary.

1.9.2 DeInit Functions

No specific handling necessary.

1.9.3 Scheduled Functions

There are no scheduled functions.

1.9.4 Crc Dependency

As E2E requires Crc functions, the Crc module needs to be integrated at the same time as E2E. If an external Crc library is used, the required interfaces and functionality must be provided as this check is not done in E2E.

1.9.5 E2E/Crc Interaction (E2E Services Stack)

In the E2E/Crc toolchain there is no private component available, so both components have to be available. Crc and E2E use the data type definitions, basic macros and memory layout provided via the central header files like Platform_-Types.h

If an external Crc library is used, the AUTOSAR interfaces must be provided in the module header Crc.h, which has to be included along with the module header E2E.h

The Crc module must be configured in the ECU Configuration Navigator.

1.9.6 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The E2E module does not use any exclusive OS resources.

1.10 Hardware Requirements

Information relating to the hardware resources required by this module is not currently available.

2 End to End Library Transformer

Module name	E2EXf
Package	RTA-SAFE
AUTOSAR Module ID	176
AUTOSAR Version	22-11

2.1 Introduction

The End to End Library Transformer module (E2EXf), which was introduced by Autosar 4.2.1, is in the system services abstraction layer, forming part of the safety (RTA-SAFE) stack.

The diagram below shows the position of E2EXf in the AUTOSAR layered architecture.

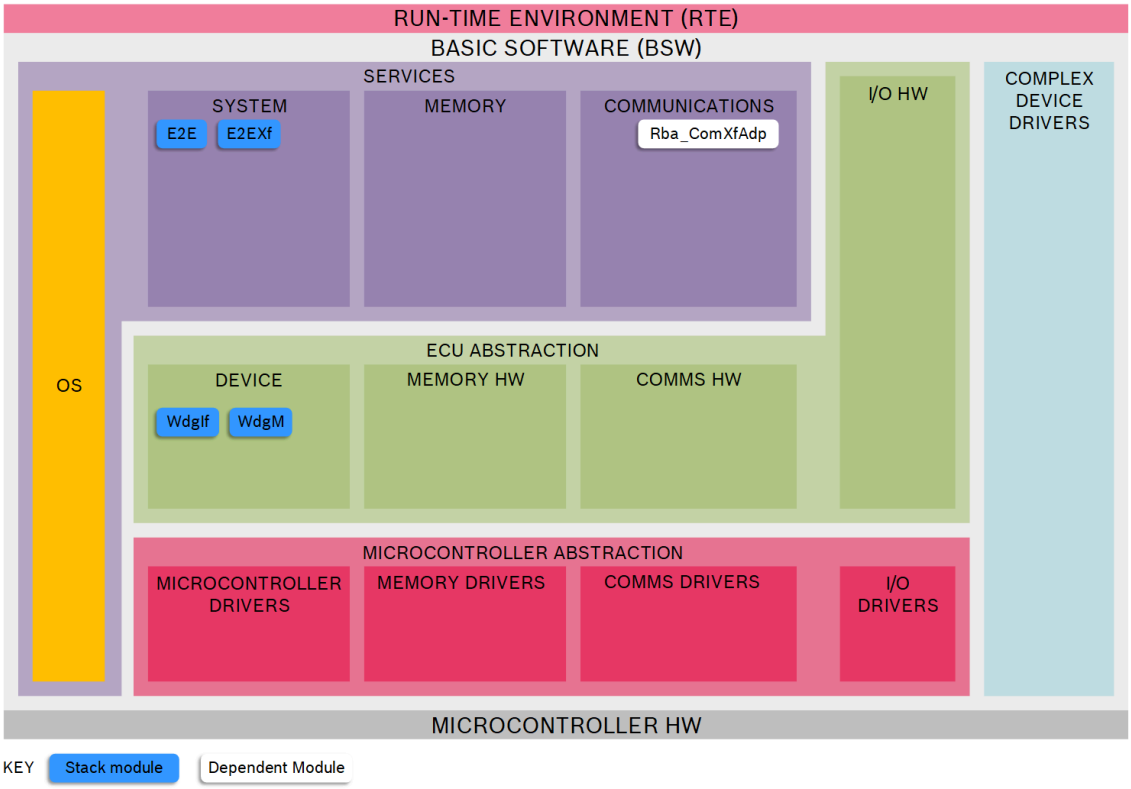

Figure 2.1.RTA-SAFE Architecture

E2EXf provides a new standardized way to invoke the End-To-End Communication Protection Library (E2E) and is responsible for protecting safety-related data elements. It encapsulates the complexity of configuring and handling of the E2E Library by offering a standard Transformer interface, thus negating the need for its caller to know the E2E internals.

E2E Protection through the E2EXf is performed on the serialized representation of data elements, on the same bit layout as the one transmitted on the bus. The

serialization is performed by a transformer above E2EXf.

- On the sender side, E2EXf "E2E-protects" the data.
- On the receiver side, E2EXf "E2E-checks" the data, providing the result of the checks through XfrmAdp to the SWC.

If a receiving SWC doesn't read the E2EXf return codes, it is fully transparent to the communicating SWCs whether or not the data is E2E-protected on the bus.

All algorithms are provided by the E2E Library (protect, check, state machine etc). E2EXf invokes the E2E Library providing the configuration and state. The routines of the E2E library protect the data exchange between software components, both for inter and intra ECU communication.

2.1.1 Crc Library

The protection of data streams are realized by complex Cyclic Redundancy Checks (provided either by the Crc module, or via an external library) over the data stream and the data identifier. The data identifier is an identifier for the sender, and is responsible for the sender-receiver authentication. Crc checks can be called manually from the ASW or BSW, or called automatically from E2E.

Module Configuration

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

2.2 Supported Features

2.2.1 General

- E2EXf simplifies the handling of the E2E protection routines and performs related data transformation.
- It provides the data structures required by E2E and is responsible for the correct invocation of the E2E library routines, as well as pre-processing and post-processing the data between E2E protection operations.
- E2EXf ensures a correct communication of I-signals through QM communication stack.
- E2EXf instantiates the E2E configuration and E2E state data structures based on its configuration (all E2E profiles may be used to protect data).
- Supports In-place and Out-of-Place processing.

2.2.2 Transformer Adaptor (XfrmAdp)

The transformer functionality described by AUTOSAR includes the manipulation of serialized data before handing it over to the next transformer or SWC.

The diagram below shows the standard AUTOSAR approach, where the RTE is responsible for allocating the memory necessary for the data transformation, and in the case of multiple data transformations, for chaining the different transformers.

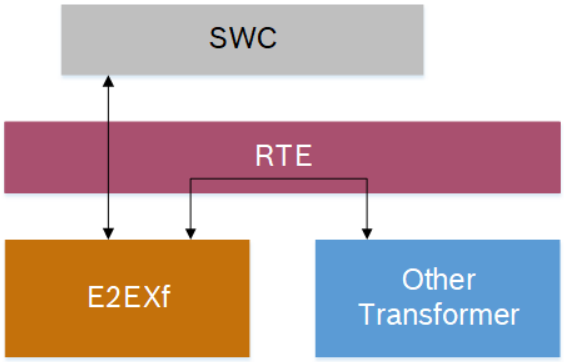

Figure 2.2.Architecture Overview

Because the RTE doesn't currently support the use of transformers, an RTA-BSW workaround makes use of a Transformer Adaptor, which is responsible for allocating the memory needed for a transformation, and to call the transformer functions. From the RTE point of view, all communication that requires data transformation is remodelled as local communication to the Adaptor Complex Driver (XfrmAdp), as shown in this second diagram.

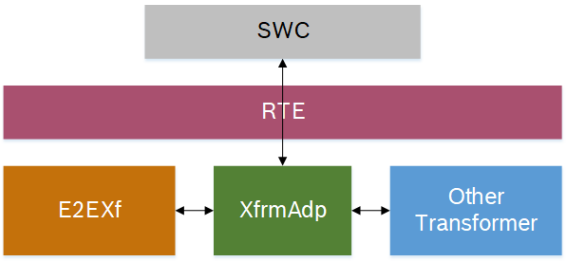

Figure 2.3.Architecture Overview with Adaptor

2.3 Extensions

2.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

2.3.1.1 E2EXf

Short Name	E2EXf
Description	RB specific configuration of the E2EXf (E2E transformer) module.
Multiplicity	0..1
Default Value	-
Type	ECUC-MODULE-DEF

2.3.1.2 E2EXf/E2EXfRb_General

Short Name	E2EXfRb_General
Long Name	E2EXf general configurations
Description	General configuration of component E2EXf. The parameters contain defines for the component behaviour and provided APIs.
Multiplicity	1..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.3 E2EXf/E2EXfRb_General/E2EXfRb_Variant

Short Name	E2EXfRb_Variant
Long Name	Variant selection
Description	With this parameter the implementation variant can be selected. Currently only a single variant implemented so this parameter has no effect on the generated code.
Multiplicity	1..1
Default Value	E2EXF_STANDARD_V01
Enumeration Literals	E2EXF_STANDARD_V01
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.4 E2EXf/E2EXfRb_General/rba_E2EXfrmBufferLengthType

Short Name	rba_E2EXfrmBufferLengthType
Description	With this parameter, datatype of the bufferLength parameter can be defined
Multiplicity	0..1
Default Value	uint32
Enumeration Literals	uint16, uint32
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.5 E2EXf/E2EXfRb_Transformer

Short Name	E2EXfRb_Transformer
Long Name	E2EXf transformer configuration
Description	E2EXf transformer configuration parameters
Multiplicity	0..*
Default Value	-

Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.6 E2EXf/E2EXfRb_Transformer/E2EXfRb_CSSignalType

Short Name	E2EXfRb_CSSignalType
Long Name	CSSignalType
Description	Describes the Client Server signal type.
Multiplicity	0..1
Default Value	CallSignal
Enumeration Literals	CallSignal, ReturnSignal
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.7 E2EXf/E2EXfRb_Transformer/E2EXfRb_CommunicationDirection

Short Name	E2EXfRb_CommunicationDirection
Long Name	CommunicationDirection
Description	Describes the communication direction.
Multiplicity	1..1
Default Value	-
Enumeration Literals	in, out
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.8 E2EXf/E2EXfRb_Transformer/E2EXfRb_CommunicationType

Short Name	E2EXfRb_CommunicationType
Long Name	CommunicationType
Description	Describes the communication type.
Multiplicity	1..1
Default Value	-
Enumeration Literals	SenderReceiver, ClientServer
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.9 E2EXf/E2EXfRb_Transformer/E2EXfRb_DisableEndToEndCheck

Short Name	E2EXfRb_DisableEndToEndCheck
Long Name	DisableEndToEndCheck
Description	Disables/Enables the E2E check. if True then E2E Transformer shall skip the invocation of the E2E Library - it shall only perform buffer processing
Multiplicity	0..1
Default Value	false

Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.10 E2EXf/E2EXfRb_Transformer/E2EXfRb_HeaderLength

Short Name	E2EXfRb_HeaderLength
Long Name	HeaderLength
Description	Defines the length of the header (in bits) this transformer will add in front of the data.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.11 E2EXf/E2EXfRb_Transformer/E2EXfRb_InPlace

Short Name	E2EXfRb_InPlace
Long Name	InPlace
Description	Selects the buffer handling mode.If true: the transformer uses the input buffer as output buffer.If false: the transformer uses separate buffer for the input and output.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.12 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration

Short Name	E2EXfRb_ProfileConfiguration
Long Name	Profile configuration
Description	Select profile type by choosing one of the configuration container
Multiplicity	1..1
Default Value	-
Type	ECUC-CHOICE-CONTAINER-DEF

2.3.1.13 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile01

Short Name	E2EXfRb_GM_Profile01
Long Name	GM Profile01
Description	GM Custom Profile 1 configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.14 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile01/E2EXfRb_CounterLength

Short Name	E2EXfRb_CounterLength
Long Name	CounterLength
Description	Length of Active Rolling Counter as 2, 4 or 8 bits
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.15 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile01/E2EXfRb_CounterOffset

Short Name	E2EXfRb_CounterOffset
Long Name	CounterOffset
Description	Offset of the counter in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.16 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile01/E2EXfRb_DataLength

Short Name	E2EXfRb_DataLength
Long Name	DataLength
Description	Length of Data in bits
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.17 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile01/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.18 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile01/E2EXfRb_MaxDeltaCounterInit

Short Name	E2EXfRb_MaxDeltaCounterInit
Long Name	MaxDeltaCounterInit
Description	Initial maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.19 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile02

Short Name	E2EXfRb_GM_Profile02
Long Name	GM Profile02
Description	GM Custom Profile 2 configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.20 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile02/E2EXfRb_ChecksumOffset

Short Name	E2EXfRb_ChecksumOffset
Long Name	ChecksumOffset
Description	Offset of the Checksum in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.21 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile02/E2EXfRb_CounterOffset

Short Name	E2EXfRb_CounterOffset
Long Name	CounterOffset
Description	Offset of the counter in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.22 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile02/E2EXfRb_DataID

Short Name	E2EXfRb_DataID
Long Name	DataID
Description	Data ID for the given message which is used for Checksum computation
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.23 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile02/E2EXfRb_DataLength

Short Name	E2EXfRb_DataLength
Long Name	DataLength
Description	Length of Data in bits
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.24 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile02/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.25 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile02/E2EXfRb_MaxDeltaCounterInit

Short Name	E2EXfRb_MaxDeltaCounterInit
Long Name	MaxDeltaCounterInit
Description	Initial maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.26 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile03

Short Name	E2EXfRb_GM_Profile03
Long Name	GM Profile03
Description	GM Custom Profile 3 configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.27 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile03/E2EXfRb_CounterOffset

Short Name	E2EXfRb_CounterOffset
Long Name	CounterOffset
Description	Offset of the counter in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.28 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile03/E2EXfRb_CrcOffset

Short Name	E2EXfRb_CrcOffset
Long Name	CrcOffset
Description	Offset of the CRC in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.29 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile03/E2EXfRb_DataID

Short Name	E2EXfRb_DataID
Long Name	DataID
Description	Data ID for the given message which is used for CRC computation
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.30 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/

E2EXfRb_GM_Profile03/E2EXfRb_DataLength

Short Name	E2EXfRb_DataLength
Long Name	DataLength
Description	Length of Data in bits
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.31 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile03/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.32 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile03/E2EXfRb_MaxDeltaCounterInit

Short Name	E2EXfRb_MaxDeltaCounterInit
Long Name	MaxDeltaCounterInit
Description	Initial maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.33 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile04

Short Name	E2EXfRb_GM_Profile04
Long Name	GM Profile04
Description	GM Custom Profile 4 configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.34 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile04/E2EXfRb_CounterOffset

Short Name	E2EXfRb_CounterOffset
Long Name	CounterOffset
Description	Offset of the counter in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.35 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile04/E2EXfRb_DataLength

Short Name	E2EXfRb_DataLength
Long Name	DataLength
Description	Length of Data in bits
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.36 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile04/E2EXfRb_EndBitPos

Short Name	E2EXfRb_EndBitPos
Long Name	EndBitPos
Description	End Bit Position of the last Signal in the Signal Group in Big Endian format.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.37 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile04/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.38 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile04/E2EXfRb_MaxDeltaCounterInit

Short Name	E2EXfRb_MaxDeltaCounterInit
Long Name	MaxDeltaCounterInit
Description	Initial maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.39 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile04/E2EXfRb_PVOffset

Short Name	E2EXfRb_PVOffset
Long Name	Protection value Offset
Description	Offset of the Protection value in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.40 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile04/E2EXfRb_PayloadSize

Short Name	E2EXfRb_PayloadSize
Long Name	Payload Size
Description	Size of Actual Payload in the Data array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.41 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_GM_Profile04/E2EXfRb_StartBitPos

Short Name	E2EXfRb_StartBitPos
Long Name	StartBitPos
Description	Start Bit Position of the first Signal in the Signal Group in Big Endian format.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.42 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile1

Short Name	E2EXfRb_Profile1
Long Name	Profile 1
Description	Profile 1 configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.43 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile1/E2EXfRb_CounterOffset

Short Name	E2EXfRb_CounterOffset
Long Name	Counter Offset
Description	Offset of the counter in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.44 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile1/E2EXfRb_CrcOffset

Short Name	E2EXfRb_CrcOffset
Long Name	CRC Offset
Description	Offset of the CRC in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.45 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile1/E2EXfRb_DataID

Short Name	E2EXfRb_DataID
Long Name	DataId

Description	This represents a unique numerical identifier. Note: ID is used for protection against masquerading. The details concerning the maximum number of values (this information is specific for each E2E profile) applicable for this attribute are controlled by a semantic constraint that depends on the category of the EndToEndProtection.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.46 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile1/E2EXfRb_DataIdMode

Short Name	E2EXfRb_DataIdMode
Long Name	Data ID Mode
Description	This attribute describes the inclusion mode that is used to include the implicit two-byte Data ID in the one-byte CRC.
Multiplicity	1..1
Default Value	-
Enumeration Literals	E2E_P01_DATAID_BOTH, E2E_P01_DATAID_ALT, E2E_P01_DATAID_ LOW, E2E_P01_DATAID_NIBBLE
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.47 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile1/E2EXfRb_DataIdNibbleOffset

Short Name	E2EXfRb_DataIdNibbleOffset
Long Name	Data ID Nibble Offset
Description	Offset of the Data ID nibble in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.48 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile1/E2EXfRb_DataLength

Short Name	E2EXfRb_DataLength
Long Name	DataLength
Description	Length of Data in bits
Multiplicity	1..1
Default Value	-

Type	ECUC-INTEGER-PARAM-DEF

2.3.1.49 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile1/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounterInit
Description	Initial maximum allowed gap between two counter values of two consecutively received valid Data.
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.50 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile1/E2EXfRb_MaxNoNewOrRepeatedData

Short Name	E2EXfRb_MaxNoNewOrRepeatedData
Long Name	MaxNoNewOrRepeatedData
Description	The maximum allowed amount of consecutive failed counter checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.51 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile1/E2EXfRb_ProfileBehavior

Short Name	E2EXfRb_ProfileBehavior
Long Name	ProfileBehavior
Description	Behavior of the check functionality (Only valid for Profile1 and Profile2).This parameter is provided to select the profile behaviour. This is an enumeration parameter. It contains two values : R4_2 and PRE_R4_2. If selected R4_2, the SM behaves according to AR42 E2E SWS. If selected PRE_R4_2, the SM behavior is prior to AR42.
Multiplicity	0..1
Default Value	R4_2
Enumeration Literals	PRE_R4_2, R4_2
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.52 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile1/E2EXfRb_SyncCounterInit

Short Name	E2EXfRb_SyncCounterInit
Long Name	SyncCounterInit
Description	Number of checks required for validating the consistency of the counter that must be received with a valid counter (i.e. counter within the allowed lock-in range) after the detection of an unexpected behavior of a received counter
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.53 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile11

Short Name	E2EXfRb_Profile11
Long Name	Profile 11
Description	Profile 11 configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.54 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile11/E2EXfRb_CounterOffset

Short Name	E2EXfRb_CounterOffset
Long Name	Counter Offset
Description	Offset of the counter in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.55 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile11/E2EXfRb_CrcOffset

Short Name	E2EXfRb_CrcOffset
Long Name	CRC Offset
Description	Offset of the CRC in the Data[] array in bits.
Multiplicity	1..1

Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.56 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile11/E2EXfRb_DataID

Short Name	E2EXfRb_DataID
Long Name	DataId
Description	This represents a unique numerical identifier. Note: ID is used for protection against masquerading. The details concerning the maximum number of values (this information is specific for each E2E profile) applicable for this attribute are controlled by a semantic constraint that depends on the category of the EndToEndProtection.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.57 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile11/E2EXfRb_DataIdMode

Short Name	E2EXfRb_DataIdMode
Long Name	Data ID Mode
Description	This attribute describes the inclusion mode that is used to include the implicit two-byte Data ID in the one-byte CRC.
Multiplicity	1..1
Default Value	-
Enumeration Literals	E2E_P11_DATAID_BOTH, E2E_P11_DATAID_NIBBLE
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.58 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile11/E2EXfRb_DataIdNibbleOffset

Short Name	E2EXfRb_DataIdNibbleOffset
Long Name	Data ID Nibble Offset
Description	Offset of the Data ID nibble in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.59 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile11/E2EXfRb_DataLength

Short Name	E2EXfRb_DataLength
Long Name	DataLength
Description	Length of Data in bits
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.60 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile11/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.61 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile2

Short Name	E2EXfRb_Profile2
Long Name	Profile 2
Description	Profile 2 configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.62 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile2/E2EXfRb_DataID

Short Name	E2EXfRb_DataID
Long Name	DataId
Description	This represents a unique numerical identifier. Note: ID is used for protection against masquerading. For this profile type (Profile 2) exactly 16 DataIds required to be defined.
Multiplicity	16..16
Default Value	-

Type	ECUC-INTEGER-PARAM-DEF

2.3.1.63 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile2/E2EXfRb_DataLength

Short Name	E2EXfRb_DataLength
Long Name	DataLength
Description	Length of Data in bits
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.64 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile2/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounterInit
Description	Initial maximum allowed gap between two counter values of two consecutively received valid Data.
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.65 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile2/E2EXfRb_MaxNoNewOrRepeatedData

Short Name	E2EXfRb_MaxNoNewOrRepeatedData
Long Name	MaxNoNewOrRepeatedData
Description	The maximum allowed amount of consecutive failed counter checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.66 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile2/E2EXfRb_Offset

Short Name	E2EXfRb_Offset
Long Name	Offset
Description	Offset of the E2E header in the Data[] array in bits.
Multiplicity	1..1
Default Value	-

Type	ECUC-INTEGER-PARAM-DEF

2.3.1.67 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile2/E2EXfRb_ProfileBehavior

Short Name	E2EXfRb_ProfileBehavior
Long Name	ProfileBehavior
Description	Behavior of the check functionality (Only valid for Profile1 and Profile2).This parameter is provided to select the profile behaviour. This is an enumeration parameter. It contains two values : R4_2 and PRE_R4_2. If selected R4_2, the SM behaves according to AR42 E2E SWS. If selected PRE_R4_2, the SM behavior is prior to AR42.
Multiplicity	0..1
Default Value	R4_2
Enumeration Literals	PRE_R4_2, R4_2
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.68 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile2/E2EXfRb_SyncCounterInit

Short Name	E2EXfRb_SyncCounterInit
Long Name	SyncCounterInit
Description	Number of checks required for validating the consistency of the counter that must be received with a valid counter (i.e. counter within the allowed lock-in range) after the detection of an unexpected behavior of a received counter
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.69 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile22

Short Name	E2EXfRb_Profile22
Long Name	Profile 22
Description	Profile 22 configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.70 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile22/E2EXfRb_DataID

Short Name	E2EXfRb_DataID
Long Name	DataId
Description	This represents a unique numerical identifier. Note: ID is used for protection against masquerading. The details concerning the maximum number of values (this information is specific for each E2E profile) applicable for this attribute are controlled by a semantic constraint that depends on the category of the EndToEndProtection.
Multiplicity	16..16
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.71 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile22/E2EXfRb_DataLength

Short Name	E2EXfRb_DataLength
Long Name	DataLength
Description	Length of Data in bits
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.72 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile22/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.73 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile22/E2EXfRb_Offset

Short Name	E2EXfRb_Offset
Long Name	Offset

Description	Offset of the E2E header in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.74 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile4

Short Name	E2EXfRb_Profile4
Long Name	Profile 4
Description	Profile 4 configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.75 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile4/E2EXfRb_DataID

Short Name	E2EXfRb_DataID
Long Name	DataId
Description	This represents a unique numerical identifier. Note: ID is used for protection against masquerading. The details concerning the maximum number of values (this information is specific for each E2E profile) applicable for this attribute are controlled by a semantic constraint that depends on the category of the EndToEndProtection.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.76 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile4/E2EXfRb_MaxDataLength

Short Name	E2EXfRb_MaxDataLength
Long Name	MaxDataLength
Description	Maximum length of Data in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.77 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/

E2EXfRb_Profile4/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.78 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile4/E2EXfRb_MinDataLength

Short Name	E2EXfRb_MinDataLength
Long Name	MinDataLength
Description	Minimum length of Data in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.79 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile4/E2EXfRb_Offset

Short Name	E2EXfRb_Offset
Long Name	Offset
Description	Offset of the E2E header in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.80 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile44

Short Name	E2EXfRb_Profile44
Long Name	Profile 44
Description	Profile 44 configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.81 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/

E2EXfRb_Profile44/E2EXfRb_DataID

Short Name	E2EXfRb_DataID
Long Name	DataId
Description	This represents a unique numerical identifier. Note: ID is used for protection against masquerading. The details concerning the maximum number of values (this information is specific for each E2E profile) applicable for this attribute are controlled by a semantic constraint that depends on the category of the EndToEndProtection.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.82 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile44/E2EXfRb_MaxDataLength

Short Name	E2EXfRb_MaxDataLength
Long Name	MaxDataLength
Description	Maximum length of Data in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.83 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile44/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.84 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile44/E2EXfRb_MinDataLength

Short Name	E2EXfRb_MinDataLength
Long Name	MinDataLength
Description	Minimum length of Data in bits.

Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.85 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile44/E2EXfRb_Offset

Short Name	E2EXfRb_Offset
Long Name	Offset
Description	Offset of the E2E header in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.86 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile4m

Short Name	E2EXfRb_Profile4m
Long Name	Profile 4m
Description	Profile 4m configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.87 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile4m/E2EXfRb_DataID

Short Name	E2EXfRb_DataID
Long Name	DataId
Description	This represents a unique numerical identifier. Note: ID is used for protection against masquerading. The details concerning the maximum number of values (this information is specific for each E2E profile) applicable for this attribute are controlled by a semantic constraint that depends on the category of the EndToEndProtection.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.88 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile4m/E2EXfRb_MaxDataLength

Short Name	E2EXfRb_MaxDataLength
Long Name	MaxDataLength
Description	Maximum length of Data in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.89 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile4m/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.90 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile4m/E2EXfRb_MinDataLength

Short Name	E2EXfRb_MinDataLength
Long Name	MinDataLength
Description	Minimum length of Data in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.91 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile4m/E2EXfRb_Offset

Short Name	E2EXfRb_Offset
Long Name	Offset
Description	Offset of the E2E header in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.92 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile4m/E2EXfRb_SourceID

Short Name	E2EXfRb_SourceID
Long Name	SourceId
Description	SourceID is a unique identifier of the source of the exchanged data/information. In case C/S (method) communication, each single client gets a separate sourceID.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.93 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile5

Short Name	E2EXfRb_Profile5
Long Name	Profile 5
Description	Profile 5 configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.94 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile5/E2EXfRb_DataID

Short Name	E2EXfRb_DataID
Long Name	DataId
Description	This represents a unique numerical identifier. Note: ID is used for protection against masquerading. The details concerning the maximum number of values (this information is specific for each E2E profile) applicable for this attribute are controlled by a semantic constraint that depends on the category of the EndToEndProtection.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.95 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile5/E2EXfRb_DataLength

Short Name	E2EXfRb_DataLength

Long Name	DataLength
Description	Length of Data in bits
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.96 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/E2EXfRb_Profile5/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.97 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/E2EXfRb_Profile5/E2EXfRb_Offset

Short Name	E2EXfRb_Offset
Long Name	Offset
Description	Offset of the E2E header in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.98 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/E2EXfRb_Profile6

Short Name	E2EXfRb_Profile6
Long Name	Profile 6
Description	Profile 6 configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.99 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/E2EXfRb_Profile6/E2EXfRb_DataID

Short Name	E2EXfRb_DataID

Long Name	DataId
Description	This represents a unique numerical identifier. Note: ID is used for protection against masquerading. The details concerning the maximum number of values (this information is specific for each E2E profile) applicable for this attribute are controlled by a semantic constraint that depends on the category of the EndToEndProtection.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.100 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile6/E2EXfRb_MaxDataLength

Short Name	E2EXfRb_MaxDataLength
Long Name	MaxDataLength
Description	Maximum length of Data in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.101 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile6/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.102 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile6/E2EXfRb_MinDataLength

Short Name	E2EXfRb_MinDataLength
Long Name	MinDataLength
Description	Minimum length of Data in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.103 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile6/E2EXfRb_Offset

Short Name	E2EXfRb_Offset
Long Name	Offset
Description	Offset of the E2E header in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.104 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile7

Short Name	E2EXfRb_Profile7
Long Name	Profile 7
Description	Profile 7 configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.105 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile7/E2EXfRb_DataID

Short Name	E2EXfRb_DataID
Long Name	DataId
Description	This represents a unique numerical identifier. Note: ID is used for protection against masquerading. The details concerning the maximum number of values (this information is specific for each E2E profile) applicable for this attribute are controlled by a semantic constraint that depends on the category of the EndToEndProtection.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.106 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile7/E2EXfRb_MaxDataLength

Short Name	E2EXfRb_MaxDataLength
Long Name	MaxDataLength

Description	Maximum length of Data in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.107 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile7/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.108 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile7/E2EXfRb_MinDataLength

Short Name	E2EXfRb_MinDataLength
Long Name	MinDataLength
Description	Minimum length of Data in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.109 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile7/E2EXfRb_Offset

Short Name	E2EXfRb_Offset
Long Name	Offset
Description	Offset of the E2E header in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.110 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile7m

Short Name	E2EXfRb_Profile7m
Long Name	Profile 7m

Description	Profile 7m configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.111 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile7m/E2EXfRb_DataID

Short Name	E2EXfRb_DataID
Long Name	DataId
Description	This represents a unique numerical identifier. Note: ID is used for protection against masquerading. The details concerning the maximum number of values (this information is specific for each E2E profile) applicable for this attribute are controlled by a semantic constraint that depends on the category of the EndToEndProtection.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.112 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile7m/E2EXfRb_MaxDataLength

Short Name	E2EXfRb_MaxDataLength
Long Name	MaxDataLength
Description	Maximum length of Data in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.113 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile7m/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.114 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile7m/E2EXfRb_MinDataLength

Short Name	E2EXfRb_MinDataLength
Long Name	MinDataLength
Description	Minimum length of Data in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.115 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile7m/E2EXfRb_Offset

Short Name	E2EXfRb_Offset
Long Name	Offset
Description	Offset of the E2E header in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.116 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile7m/E2EXfRb_SourceID

Short Name	E2EXfRb_SourceID
Long Name	SourceId
Description	SourceID is a unique identifier of the source of the exchanged data/information. In case C/S (method) communication, each single client gets a separate sourceID.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.117 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile8

Short Name	E2EXfRb_Profile8
Long Name	Profile 8
Description	Profile 8 configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.118 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile8/E2EXfRb_DataID

Short Name	E2EXfRb_DataID
Long Name	DataId
Description	This represents a unique numerical identifier. Note: ID is used for protection against masquerading. The details concerning the maximum number of values (this information is specific for each E2E profile) applicable for this attribute are controlled by a semantic constraint that depends on the category of the EndToEndProtection.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.119 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile8/E2EXfRb_MaxDataLength

Short Name	E2EXfRb_MaxDataLength
Long Name	MaxDataLength
Description	Maximum length of Data in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.120 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile8/E2EXfRb_MaxDeltaCounter

Short Name	E2EXfRb_MaxDeltaCounter
Long Name	MaxDeltaCounter
Description	Maximum allowed gap between two counter values of two consecutive checks
Multiplicity	0..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.121 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/ E2EXfRb_Profile8/E2EXfRb_MinDataLength

Short Name	E2EXfRb_MinDataLength
Long Name	MinDataLength

Description	Minimum length of Data in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.122 E2EXf/E2EXfRb_Transformer/E2EXfRb_ProfileConfiguration/E2EXfRb_Profile8/E2EXfRb_Offset

Short Name	E2EXfRb_Offset
Long Name	Offset
Description	Offset of the Data ID nibble in the Data[] array in bits.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.123 E2EXf/E2EXfRb_Transformer/E2EXfRb_StateMachineConfiguration

Short Name	E2EXfRb_StateMachineConfiguration
Long Name	State machine configuration
Description	State machine configuration
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

2.3.1.124 E2EXf/E2EXfRb_Transformer/E2EXfRb_StateMachineConfiguration/E2EXfRb_ClearToInvalid

Short Name	E2EXfRb_ClearToInvalid
Long Name	ClearToInvalid
Description	Clear monitoring window data on transition to state INVALID.
Multiplicity	0..1
Default Value	FALSE
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.125 E2EXf/E2EXfRb_Transformer/E2EXfRb_StateMachineConfiguration/E2EXfRb_MaxErrorStateInit

Short Name	E2EXfRb_MaxErrorStateInit
Long Name	MaxErrorStateInit

Description	Maximal number of checks in which ProfileStatus equal to E2E_P_ERROR was determined, within the last WindowSize checks, for the state E2E_SM_INIT.
Multiplicity	1..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.126 E2EXf/E2EXfRb_Transformer/E2EXfRb_StateMachineConfiguration/E2EXfRb_MaxErrorStateInvalid

Short Name	E2EXfRb_MaxErrorStateInvalid
Long Name	MaxErrorStateInvalid
Description	Maximal number of checks in which ProfileStatus equal to E2E_P_ERROR was determined, within the last WindowSize checks, for the state E2E_SM_INVALID.
Multiplicity	1..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.127 E2EXf/E2EXfRb_Transformer/E2EXfRb_StateMachineConfiguration/E2EXfRb_MaxErrorStateValid

Short Name	E2EXfRb_MaxErrorStateValid
Long Name	MaxErrorStateValid
Description	Maximal number of checks in which ProfileStatus equal to E2E_P_ERROR was determined, within the last WindowSize checks, for the state E2E_SM_VALID.
Multiplicity	1..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.128 E2EXf/E2EXfRb_Transformer/E2EXfRb_StateMachineConfiguration/E2EXfRb_MinOkStateInit

Short Name	E2EXfRb_MinOkStateInit
Long Name	MinOkStateInit

Description	Minimal number of checks in which ProfileStatus equal to E2E_P_OK was determined, within the last WindowSize checks, for the state E2E_SM_INIT.
Multiplicity	1..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.129 E2EXf/E2EXfRb_Transformer/E2EXfRb_StateMachineConfiguration/E2EXfRb_MinOkStateInvalid

Short Name	E2EXfRb_MinOkStateInvalid
Long Name	MinOkStateInvalid
Description	Minimal number of checks in which ProfileStatus equal to E2E_P_OK was determined, within the last WindowSize checks, for the state E2E_SM_INVALID.
Multiplicity	1..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.130 E2EXf/E2EXfRb_Transformer/E2EXfRb_StateMachineConfiguration/E2EXfRb_MinOkStateValid

Short Name	E2EXfRb_MinOkStateValid
Long Name	MinOkStateValid
Description	Minimal number of checks in which ProfileStatus equal to E2E_P_OK was determined, within the last WindowSize checks, for the state E2E_SM_VALID.
Multiplicity	1..1
Minimum Value	0
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.131 E2EXf/E2EXfRb_Transformer/E2EXfRb_StateMachineConfiguration/E2EXfRb_ProfileBehavior

Short Name	E2EXfRb_ProfileBehavior
Long Name	ProfileBehavior

Description	Behavior of the check functionality (Only valid for Profile1 and Profile2).This parameter is provided to select the profile behaviour. This is an enumeration parameter. It contains two values : R4_2 and PRE_R4_2. If selected R4_2, the SM behaves according to AR42 E2E SWS. If selected PRE_R4_2, the SM behavior is prior to AR42.
Multiplicity	0..1
Default Value	R4_2
Enumeration Literals	PRE_R4_2, R4_2
Type	ECUC-ENUMERATION-PARAM-DEF

2.3.1.132 E2EXf/E2EXfRb_Transformer/E2EXfRb_StateMachineConfiguration/E2EXfRb_WindowSize

Short Name	E2EXfRb_WindowSize
Long Name	WindowSize
Description	Size of the monitoring window for the E2E state machine
Multiplicity	0..1
Minimum Value	1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.133 E2EXf/E2EXfRb_Transformer/E2EXfRb_StateMachineConfiguration/E2EXfRb_WindowSizeInit

Short Name	E2EXfRb_WindowSizeInit
Long Name	WindowSizeInit
Description	Size of the monitoring windows for the state machine during state INIT
Multiplicity	0..1
Minimum Value	1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.134 E2EXf/E2EXfRb_Transformer/E2EXfRb_StateMachineConfiguration/E2EXfRb_WindowSizeInvalid

Short Name	E2EXfRb_WindowSizeInvalid
Long Name	WindowSizeInvalid
Description	Size of the monitoring window for the state machine during state INVALID.
Multiplicity	0..1

Minimum Value	1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.135 E2EXf/E2EXfRb_Transformer/E2EXfRb_StateMachineConfiguration/E2EXfRb_WindowSizeValid

Short Name	E2EXfRb_WindowSizeValid
Long Name	WindowSizeValid
Description	Size of the monitoring window for the state machine during state VALID.
Multiplicity	0..1
Minimum Value	1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.1.136 E2EXf/E2EXfRb_Transformer/E2EXfRb_StateMachineConfiguration/E2EXfRb_transitToInvalidExtended

Short Name	E2EXfRb_transitToInvalidExtended
Long Name	transitToInvalidExtended
Description	Restrict/allow tranistion from states INIT/NODATA to INVALID state
Multiplicity	0..1
Default Value	FALSE
Type	ECUC-BOOLEAN-PARAM-DEF

2.3.1.137 E2EXf/E2EXfRb_Transformer/E2EXfRb_UpperHeaderBitsToShift

Short Name	E2EXfRb_UpperHeaderBitsToShift
Long Name	Upper Header Bits To Shift
Description	This attribute describes the number of upper-header bits to be shifted.
Multiplicity	1..1
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

2.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

- There are no additional API defined in this version.

2.4 Limitations

2.4.1 RTE Connection

- E2EXf does not currently support a direct RTE connection (see Supported Features section for more information on the RTA-BSW workaround).

2.4.2 AUTOSAR Specification Issues

The following limitations/deviations are a result of issues identified within the AUTOSAR specification.

2.4.2.1 Incorrect API specification for E2EXf transformer functions

AUTOSAR requirements SWS_E2EXf_00032 and SWS_E2EXf_00034 imply that the inputBuffer parameter is always present, but it is optional, and only available when out-of-place transformation buffer handling is used. The description of inputBuffer has been updated to make it clear that it is an optional parameter.

2.4.2.2 Incorrect E2E unused bit masking handling

The (draft state) AUTOSAR specification for AR 23-11 states that, for E2E Profiles 2 and 22, the counter values should be placed in the E2E header from bits 12 to 15:

Byte/Bit	0								1							
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	E2E CRC												Counter			

However, the AUTOSAR specification for E2EXf states that, for profiles 2 and 22, the bits starting from Offset + 12 need to be masked:

> If (((PXX = 01 &&dataIDMode != nibble) || (PXX == 02) || (PXX = 11 &&dataIDMode != nibble) || (PXX == 22)) && BufferProperties.header Length == 16 [bits]), the function E2EXf_<transformerId>() shall, before calling E2E_PXXProtect() or E2E_PXXForward(), set 0xF in buffer at the bit offset (EndToEndTransformationDescription.crcOffset+12 for profiles P01 and P11 and EndToEndTransformationDescription.offset+12 for profiles P02 and P22).

There is therefore a deviation in the implementation, where the Masking bits start from OFFSET + 8 (instead of the specified OFFSET + 12). This is because bits 8 to 11 are vacant in E2E Profiles 2 and 22.

2.4.3 Miscellaneous

The following limitations/deviations are also present in the current implementation of E2EXf:

- E2EXf has no include dependency to Transformer headers
- PRE_R4_2 profile behaviours are also supported. Configuration parameter E2EXfRb_ProfileBehavior is used to switch between PRE_R4_2 and R4_2 state machines. This is applicable only to profile 1 and 2.By default, R4_2 is provided.
- Development Error Detect is not supported.
- Post Build Configuration mode is not supported.
- E2EXf functions are called by rba_XfrmAdp (Transformer Adapter) and not by RTE.
- An additional configuration method is provided with its own custom Parameter definition. Transformer configuration parameters are transported with a forwarder script to this component-specific configuration structure.
- When a Signal is sent/received using the LdCom module, then the transformerId is `LdCom_<ComSignalName/ComSignalGroupName>`.
- When parameters are gathered from Transformer containers, then some reference and buffer-size parameters in the containers will be automatically set using the forwarding mechanism.

2.4.4 Temporary Deviations

- E2EXf_ConfigType defined but not used.
- C/S communication supported only for m(method) profiles.
- Multiple clients from a single ECU are not supported for m(method) profiles, since single-source ID is available per signal configuration.

2.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- There are no known exclusions in this version.

2.6 Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

- There are no known deviations in this version.

2.7 API Definition

List of supported API functions

Table 2.139.E2EXf Supported API functions

`E2EXf_<transformerId>`	Protects the array/buffer to be transmitted using the in-place transformation.
`E2EXf_Inv_<transformerId>`	Checks the received data and if the data can be used by the caller then the function returns E_OK.
`E2EXf_Init`	Initializes the state of the E2E Transformer.
`E2EXf_DeInit`	Deinitializes the E2E transformer.
`E2EXf_GetVersionInfo`	Returns the version information of this module.

For further information on these functions, please refer to AUTOSAR_SWS_E2E-Transformer.pdf

2.8 Configuration Advice

NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

2.8.1 Configuration Workflow

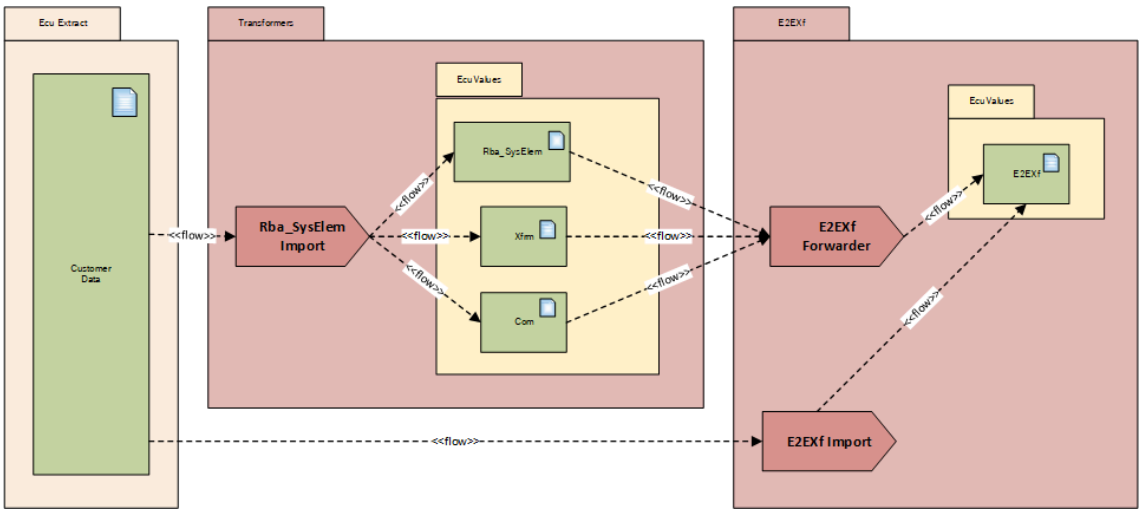

Figure 2.4.Configuration Workflow

There are three potential ways to get all the configuration parameters necessary

for E2EXf.

1. Directly configure E2EXf component in the ECU Conf Navigator using its own parameter definitions.
2. Obtain configuration parameters from other AUTOSAR components using a Forwarder script.
3. Obtain configuration parameters from an Ecu Extract.

Methods 1 and 2 are currently available, providing the flexibility of manual configuration of E2EXf, and automatic configuration by obtaining configuration parameters from other AUTOSAR components.

2.8.1.1 Configuration Parameters

AUTOSAR does not specify any configuration parameters at the E2EXf level. Instead, all of the configuration values are derived from Transformer configurations. However, as E2EXf has been developed separately from the Transformers package, it was necessary to provide configurability directly within E2EXf to ease testing and provide a simple method of configuration.

2.8.1.2 Configuration Parameter Checking

The large amount of configuration parameters used to configure E2EXf may be overwhelming for users not familiar with E2EXf and the E2E functionality in general. To ease the configuration and integration process, configuration parameter checking has been introduced. The following checks are performed on all parameters and containers:

- Check if all mandatory parameters are defined.
- Check if all required containers are defined.
- Check if parameter counts are valid (Data Id).
- Check if parameter values are valid for the given E2E Profile configuration (check offsets and Data lengths multiplicity, check Data length validity, and check if parameters are in valid value range).

2.8.1.3 Parameter Forwarding

If parameters are obtained from Transformer containers, some reference and buffer-size parameters will be automatically set using a forwarding mechanism. This operation is performed while E2EXf is gathering all relevant information from other Autosar components.

2.9 Integration Advice

2.9.1 Init Functions

E2EXf is initialized by E2EXf_Init.

2.9.2 DeInit Functions

E2EXf_DeInit de-initializes the E2EXf module.

2.9.3 Scheduled Functions

E2EXf does not have a scheduled function.

2.9.4 E2E/E2EXf/Crc Interaction (E2E Services Stack)

In the E2E/E2EXf/Crc toolchain there is no private component available, all components can be called directly by the user or are called automatically within the E2EServices package, so all three components have to be available.

All three components use the data type definitions, basic macros and memory layout provided via the central header files like Platform_Types.h.

If an external Crc library is used, the AUTOSAR interfaces must be provided in the module headers Crc.h and E2E.h, which have to be included along with the module header E2EXf.h

The Crc module must be configured in the ECU Configuration Navigator.

2.9.5 Hardware Dependencies

E2EXf is not hardware dependent, but Crc provides CRC calculations supported by specific hardware units. The selection between hardware-supported CRC calculation and plain software calculation is part of the Crc configuration.

2.9.6 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The E2EXf module does not use any exclusive OS resources.

2.10 Hardware Requirements

Information relating to the hardware resources required by this module is not currently available.

3 Watchdog Interface

Module name	Wdglf
Package	RTA-SAFE
AUTOSAR Module ID	043
AUTOSAR Version	22-11

3.1 Introduction

The Watchdog Interface (Wdglf) is in the devices ECU abstraction layer, forming part of the safety (RTA-SAFE) module stack. The diagram below shows the position of Wdglf in the AUTOSAR layered architecture.

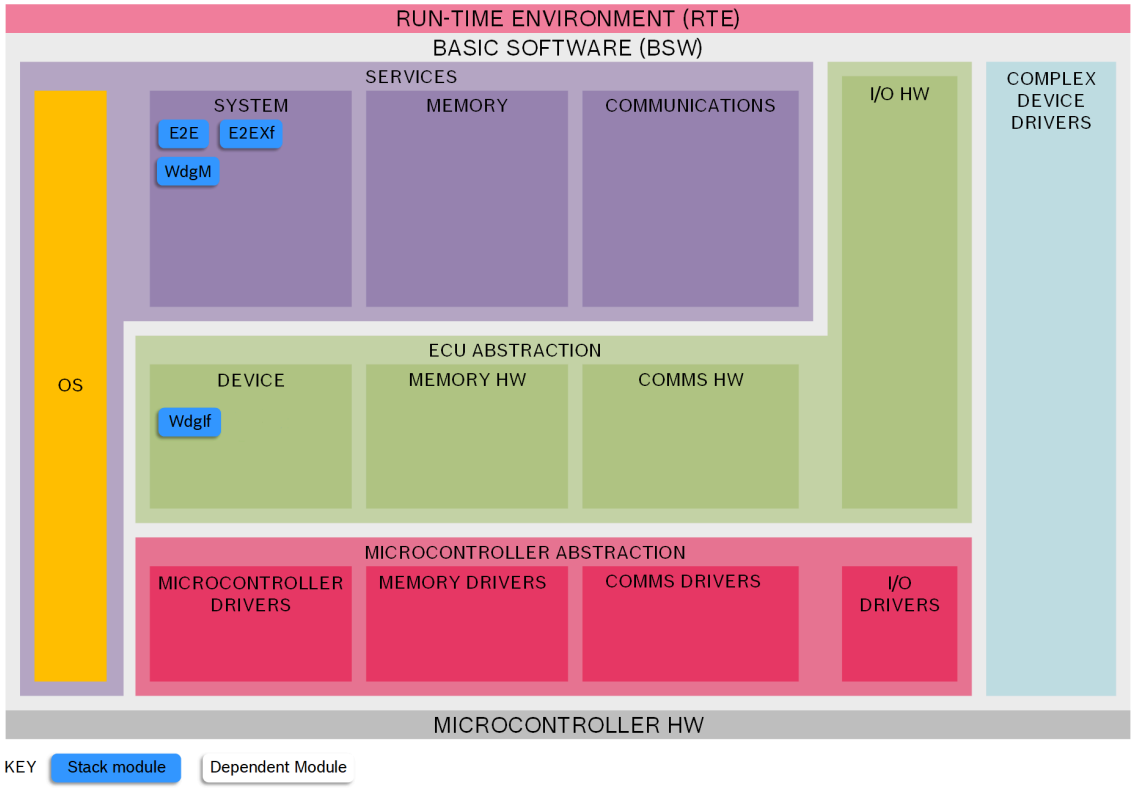

Figure 3.1.RTA-SAFE Architecture

As shown in the diagram below, Wdglf is located between the upper watchdog services layer (i.e Watchdog Manager, WdgM) and the low level watchdog device drivers.

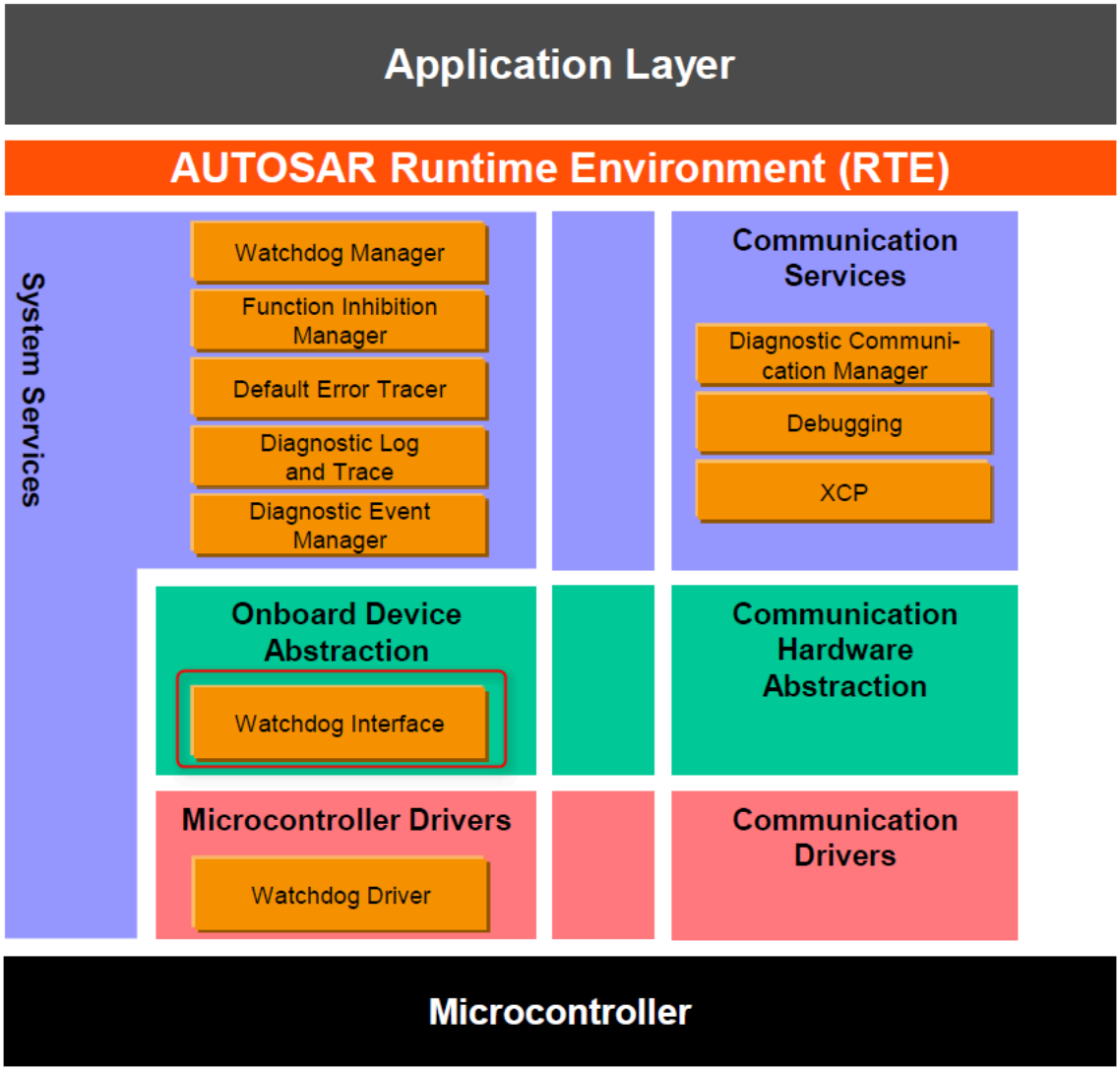

Figure 3.2.The Wdglf Module

The purpose of the Wdglf is to:

- Provide uniform access to services of the underlying watchdog drivers (e.g. mode switching and setting trigger conditions).
- Encapsulate lower layer internal or external watchdog drivers and provide a common path to access multiple watchdog drivers.
- Support AUTOSAR standard watchdog device drivers.

3.1.1 Non-AUTOSAR Extension

The RTA-BSW implementation of Wdglf includes a non-AUTOSAR extension, which provides an interface to access lower layer Bosch ASICs that have an external watchdog. See the Supported Features section for more details.

3.1.2 Dependencies/Interaction

WdgM

WdgM uses WdgIf interfaces, which in turn call the respective watchdog driver APIs to change the mode and reload the trigger condition values of the Wdg drivers.

Wdg

The references of Watchdog Drivers are provided in WdgIf to encapsulate the driver from the higher layer.

Module Configuration

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

3.2 Supported Features

3.2.1 AUTOSAR Features

WdgIf provides the below interfaces, which allow uniform access to services of the underlying watchdog drivers.

3.2.1.1 WdgIf_SetMode

WdgIf_SetMode changes the mode of the lower layer watchdog driver, which will lead to longer (Slow Mode), shorter (Fast Mode) or turn off (Off Mode) timeout periods. WdgIf_SetMode calls Wdg_<watchdog_driver_name>_SetMode(WdgMode) of the underlying watchdog driver.

3.2.1.2 WdgIf_SetTriggerCondition

WdgIf_SetTriggerCondition provides timeout period information to lower layer drivers. The lower layer driver should not timeout for the period passed by the higher. WdgIf_SetTriggerCondition calls Wdg_<watchdog_driver_name>_SetTriggerCondition(WdgTimeout) of the underlying watchdog driver.

3.2.2 Non AUTOSAR Features

In addition to the above features, WdgIf also supports a non-AUTOSAR extension that provides an interface to access lower layer Bosch-specific ASICs which have extensions to the AUTOSAR watchdog API. These non-AUTOSAR extensions have a prefix of WdgIf_Rb_*

Although the AUTOSAR features of WdgIf are hardware independent, the non-AUTOSAR features do have some dependencies. The following Bosch-specific ASICs, which have an external watchdog, are supported by WdgIf.

- rba_IoExtCy327
- rba_IoExtCy329
- rba_IoExtL9779

- rba_IoExtCy326

> **NOTE**
> Currently the BOSCH internal ASIC (which has an external watchdog) is not controlled by the Watchdog Manager module.

3.3 Extensions

3.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

3.3.1.1 WdgIf/WdgIfDevice/WdgIfRbExtWdgDriverName

Short Name	WdgIfRbExtWdgDriverName
Long Name	External Watchdog driver
Description	This is the shortname of the External Watchdog Module. Integrator should configure right short name here because same name will be used for generating the SetMode and SetTriggerCondition. For example if project want to use external driver XYZ then configure WdgIfRbExtWdgDriverName as XYZ. Also user should take care of, 1. Header file should be available for driver with same name as shortname of the External Watchdog Module - XYZ.h 2. via above Header file the following interfaces of external driver should be accessible - XYZ_SetMode and XYZ_SetTriggerCondition. NOTE: This container is only for AUTOSAR interfaces generation. There is no relationship between the extranal driver names configured under WdgIfDevice and this one.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

3.3.1.2 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport

Short Name	WdgIfRbExtWdgApiSupport
Long Name	External Watchdog Api Support

Description	This Container consists of parameters which indicate whether corresponding Api (functionality) is supported by the External watchdog module (configured at parameter "WdgIfRbExtWdgModuleName") or not. This container should not be configured by Integrator. Value for the parameters of this container will be filled(forwarded) by the External watchdog module (driver). External watchdog module should forward "External watchdog module name" as shortname for this container.
Multiplicity	0..255
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.3 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport/WdgIfRbExtWdgApiGetRegVdd5OvWda

Short Name	WdgIfRbExtWdgApiGetRegVdd5OvWda
Long Name	Api GetRegVdd5OvWda Supported
Description	Indicates whether Api to read status of VDD5OV_WDA bit is supported by External Watchdog module (driver) or not. True: GetRegVdd5OvWda Api is supported by the External Watchdog module. False: GetRegVdd5OvWda Api is not supported by the External Watchdog module. If this parameter is not configured, then it will be considered as Api not supported by the External Watchdog module (that is value 'False').
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.4 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport/WdgIfRbExtWdgApiGetSwtPosn

Short Name	WdgIfRbExtWdgApiGetSwtPosn
Long Name	Api GetSwtPosn Supported

Description	Indicates whether Api to Get Switch position is supported by External Watchdog module (driver) or not. True: GetSwtPosn Api is supported by the External Watchdog module. False: GetSwtPosn Api is not supported by the External Watchdog module. If this parameter is not configured, then it will be considered as Api not supported by the External Watchdog module (that is value 'False').
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.5 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport/WdgIfRbExtWdgApiGetUMonStatus

Short Name	WdgIfRbExtWdgApiGetUMonStatus
Long Name	Api GetUMonStatus Supported
Description	Indicates whether Api to read voltage monitoring status is supported by External Watchdog module (driver) or not. True: Api GetUMonStatus is supported by the External Watchdog module. False: Api GetUMonStatus is not supported by the External Watchdog module. If this parameter is not configured, then it will be considered as Api not supported by the External Watchdog module (that is value 'False').
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.6 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport/WdgIfRbExtWdgApiGetWdgDiag

Short Name	WdgIfRbExtWdgApiGetWdgDiag
Long Name	Api GetWdgDiag Supported

Description	Indicates whether Api to read Watchdog Diagnostics is supported by External Watchdog module (driver) or not. True: Api GetWdgDiag is supported by the External Watchdog module. False: Api GetWdgDiag is not supported by the External Watchdog module. If this parameter is not configured, then it will be considered as Api not supported by the External Watchdog module (that is value 'False').
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.7 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport/WdgIfRbExtWdgApiGetWdgThd

Short Name	WdgIfRbExtWdgApiGetWdgThd
Long Name	Api GetWdgThd Supported
Description	Indicates whether Api to Get Wda threshold is supported by External Watchdog module (driver) or not. True: Api GetWdgThd is supported by the External Watchdog module. False: Api GetWdgThd is not supported by the External Watchdog module. If this parameter is not configured, then it will be considered as Api not supported by the External Watchdog module (that is value 'False').
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.8 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport/WdgIfRbExtWdgApiGetWdgTiWin

Short Name	WdgIfRbExtWdgApiGetWdgTiWin
Long Name	Api GetWdgTiWin Supported

Description	Indicates whether Api to Get Time Window value is supported by External Watchdog module (driver) or not. True: Api GetWdgTiWin is supported by the External Watchdog module. False: Api GetWdgTiWin is not supported by the External Watchdog module. If this parameter is not configured, then it will be considered as Api not supported by the External Watchdog module (that is value 'False').
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.9 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport/WdgIfRbExtWdgApiReqUMonStatus

Short Name	WdgIfRbExtWdgApiReqUMonStatus
Long Name	Api ReqUMonStatus Supported
Description	Indicates whether Api to request voltage monitoring status is supported by External Watchdog module (driver) or not. True: ReqUMonStatus Api is supported by the External Watchdog module. False: ReqUMonStatus Api is not supported by the External Watchdog module. If this parameter is not configured, then it will be considered as Api not supported by the External Watchdog module (that is value 'False').
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.10 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport/WdgIfRbExtWdgApiRIsWDA

Short Name	WdgIfRbExtWdgApiRIsWDA
Long Name	Api RIsWDA Supported

Description	Indicates whether Api to release FS0b is supported by External Watchdog module (driver) or not. True: RlsWDA Api is supported by the External Watchdog module. False: RlsWDA Api is not supported by the External Watchdog module. If this parameter is not configured, then it will be considered as Api not supported by the External Watchdog module (that is value 'False').
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.11 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport/WdgIfRbExtWdgApiSetOperMode

Short Name	WdgIfRbExtWdgApiSetOperMode
Long Name	Api SetOperMode Supported
Description	Indicates whether Api to Set the operating mode is supported by External Watchdog module (driver) or not. True: SetOperMode Api is supported by the External Watchdog module. False: SetOperMode Api is not supported by the External Watchdog module. If this parameter is not configured, then it will be considered as Api not supported by the External Watchdog module (that is value 'False').
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.12 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport/WdgIfRbExtWdgApiSetRegVdd5OvWda

Short Name	WdgIfRbExtWdgApiSetRegVdd5OvWda
Long Name	Api SetRegVdd5OvWda Supported

Description	Indicates whether Api to set status of VDD5OV_WDA bit is supported by External Watchdog module (driver) or not. True: SetRegVdd5OvWda Api is supported by the External Watchdog module. False: SetRegVdd5OvWda Api is not supported by the External Watchdog module. If this parameter is not configured, then it will be considered as Api not supported by the External Watchdog module (that is value 'False').
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.13 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport/WdgIfRbExtWdgApiSetSwtPosn

Short Name	WdgIfRbExtWdgApiSetSwtPosn
Long Name	Api SetSwtPosn Supported
Description	Indicates whether Api to Set Switch position is supported by External Watchdog module (driver) or not. True: SetSwtPosn Api is supported by the External Watchdog module. False: SetSwtPosn Api is not supported by the External Watchdog module. If this parameter is not configured, then it will be considered as Api not supported by the External Watchdog module (that is value 'False').
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.14 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport/WdgIfRbExtWdgApiSetWdgThd

Short Name	WdgIfRbExtWdgApiSetWdgThd
Long Name	Api SetWdgThd Supported

Description	Indicates whether Api to Set Wda threshold is supported by External Watchdog module (driver) or not. True: Api SetWdgThd is supported by the External Watchdog module. False: Api SetWdgThd is not supported by the External Watchdog module. If this parameter is not configured, then it will be considered as Api not supported by the External Watchdog module (that is value 'False').
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.15 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport/WdgIfRbExtWdgApiSetWdgTiWin

Short Name	WdgIfRbExtWdgApiSetWdgTiWin
Long Name	Api SetWdgTiWin Supported
Description	Indicates whether Api to Set Time Window value is supported by External Watchdog module (driver) or not. True: Api SetWdgTiWin is supported by the External Watchdog module. False: Api SetWdgTiWin is not supported by the External Watchdog module. If this parameter is not configured, then it will be considered as Api not supported by the External Watchdog module (that is value 'False').
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.16 WdgIf/WdgIfGeneral/WdgIfRbExtWdgApiSupport/WdgIfRbExtWdgApiSetWdgWin

Short Name	WdgIfRbExtWdgApiSetWdgWin
Long Name	Api SetWdgWin Supported

Description	Indicates whether Api to Set Window period and Window duty cycle is supported by External Watchdog module (driver) or not. True: Api SetWdgWin is supported by the External Watchdog module. False: Api SetWdgWin is not supported by the External Watchdog module. If this parameter is not configured, then it will be considered as Api not supported by the External Watchdog module (that is value 'False').
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

3.3.1.17 WdgIf/WdgIfGeneral/WdgIfRbExtWdgModuleName

Short Name	WdgIfRbExtWdgModuleName
Long Name	External Watchdog Module name
Description	This is module name of the External Watchdog Module. This parameter is to help generating the Api names to be used inside WdgIf from Io External Device controller. NOTE: This container is only for non-AUTOSAR interfaces generation as in WdgIfRbExtWdgApiSupport. There is no relationship between the extranal driver names configured under WdgIfDevice and this one. If this parameter is not configured, Vendor specific(related to RTA-BSW specific IoExtDev module) External Watchdog interfaces will not be generated. This shall be configured with exact Module name, as it should appear in the Prefix for the generated Api's.
Multiplicity	0..255
Default Value	-
Type	ECUC-STRING-PARAM-DEF

3.3.1.18 WdgIf/WdgIfRbConfigSet

Short Name	WdgIfRbConfigSet
Long Name	Configuration set for WdgIf
Description	 This Container consists of parameters that support Post-Build variation.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

3.3.1.19 WdgIf/WdgIfRbConfigSet/WdgIfRbExtWdgModuleSel

Short Name	WdgIfRbExtWdgModuleSel
Long Name	Watchdog Module Name
Description	This parameter is used for the selection of the External Watchdog module Name (WdgIfRbExtWdgModuleName) for particular Post-Build variant. This parameter should not be configured manually. Value for this parameter will be forwarded by the External Watchdog Module.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

3.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

Table 3.21.WdgIf Extension API

`WdgIf_Rb_RequestQuery`	Invokes a call to <Bosch_ASIC>_RequestQuery.
`WdgIf_Rb_SetResponseTime`	Invokes a call to <Bosch_ASIC>_SetResponseTime.
`WdgIf_Rb_GetReq`	Invokes a call to <Bosch_ASIC>_GetReq.
`WdgIf_Rb_SetResponse`	Invokes a call to <Bosch_ASIC>_SetResponse.
`WdgIf_Rb_GetSwtPosn`	Invokes a call to <Bosch_ASIC>___GetSwtPosn.
`WdgIf_Rb_SetSwtPosn`	Invokes a call to <Bosch_ASIC>___SetSwtPosn.
`WdgIf_Rb_Transmit`	Invokes a call to <Bosch_ASIC>_Transmit.
`WdgIf_Rb_WatchDogHandle`	Invokes a call to <Bosch_ASIC>_WatchDogHandle.
`WdgIf_Rb_WatchDogAnswer`	Invokes a call to <Bosch_ASIC>_WatchDogAnswer.
`WdgIf_Rb_GetRegCanTdi`	Invokes a call to <Bosch_ASIC>_GetRegCanTdi.
`WdgIf_Rb_GetRegInitWdr`	Invokes a call to <Bosch_ASIC>_GetRegInitWdr.
`WdgIf_Rb_GetRegWda`	Invokes a call to <Bosch_ASIC>_GetRegWda.
`WdgIf_Rb_GetWda`	Invokes a call to <Bosch_ASIC>_GetWda.
`WdgIf_Rb_SetRegCanTdi`	Invokes a call to <Bosch_ASIC>_SetRegCanTdi.
`WdgIf_Rb_SetRegInitWdr`	Invokes a call to <Bosch_ASIC>_SetRegInitWdr.
`WdgIf_Rb_GetReg_Tst0`	Invokes a call to <Bosch_ASIC>_GetReg_Tst0.
`WdgIf_Rb_SetReg_Tst0`	Invokes a call to <Bosch_ASIC>_SetReg_Tst0.
`WdgIf_Rb_GetReg_Tst1`	Invokes a call to <Bosch_ASIC>_GetReg_Tst1.
`WdgIf_Rb_SetReg_Tst1`	Invokes a call to <Bosch_ASIC>_SetReg_Tst1.
`WdgIf_Rb_StartWdaMonMode`	Invokes a call to <Bosch_ASIC>_StartWdaMon-Mode.

WdgIf_Rb_StopWdaMonMode	Invokes a call to <Bosch_ASIC>_StopWdaMon-Mode.
WdgIf_Rb_ReqWda	Invokes a call to <Bosch_ASIC>_ReqWda.
WdgIf_Rb_SetWdgThd	Invokes a call to <Bosch_ASIC>_SetWdgThd.
WdgIf_Rb_GetWdgThd	Invokes a call to <Bosch_ASIC>_GetWdgThd.
WdgIf_Rb_SetWdgTiWin	Invokes a call to <Bosch_ASIC>_SetWdgTiWin.
WdgIf_Rb_GetWdgTiWin	Invokes a call to <Bosch_ASIC>_GetWdgTiWin.
WdgIf_Rb_SetOperMode	Invokes a call to <Bosch_ASIC>_SetOperMode.
WdgIf_Rb_ReqUMonStatus	Invokes a call to <Bosch_ASIC>_ReqUMonStatus.
WdgIf_Rb_GetUMonStatus	Invokes a call to <Bosch_ASIC>_GetUMonStatus.
WdgIf_Rb_SetRegVdd5OvWda	Invokes a call to <Bosch_ASIC>_SetRegVd-d5OvWda.
WdgIf_Rb_GetRegVdd5OvWda	Invokes a call to <Bosch_ASIC>_GetRegVd-d5OvWda.
WdgIf_Rb_SetWdgWin	Invokes a call to <Bosch_ASIC>_SetWdgWin.
WdgIf_Rb_GetWdgDiag	Invokes a call to <Bosch_ASIC>_GetWdgDiag.

3.4 Limitations

- The BSW module description is generated for WdgIf by assuming a valid IoExtDev (if any External WatchDog Driver exits) is configured with module ID: 254.

- WdgIf is not following 'one fits all' approach i.e. this implementation is delivered as part of RTA-BSW and integration to another version of RTA-BSW is not supported. In such cases, a seperate risk analysis needs to be performed.

3.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- There are no known exclusions in this version.

3.6 Deviations

3.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

3.6.1.1 WdgIf

[RTA-BSW] Description	Configuration of the WdgIf (Watchdog Interface) module (with vendor specific enhancements of RB).
[AUTOSAR] Description	Configuration of the WdgIf (Watchdog Interface) module.

3.6.1.2 WdgIf/WdgIfDevice/WdgIfDeviceIndex

[RTA-BSW] Description	Represents the watchdog interface ID so that it can be referenced by the watchdog manager. This parameter should not be configured by Integrator. Value for this parameters will be filled(forwarded) by the Id Generate Script.
[AUTOSAR] Description	Represents the watchdog interface ID so that it can be referenced by the watchdog manager.

3.6.1.3 WdgIf/WdgIfDevice/WdgIfDriverRef

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

3.6.1.4 WdgIf/WdgIfGeneral/WdgIfDevErrorDetect

[RTA-BSW] Description	Switches the development error detection and notification on or off. true: detection and notification is enabled. false: detection and notification is disabled.
[AUTOSAR] Description	Switches the development error detection and notification on or off.
[REMOVED] Introduction	* true: detection and notification is enabled. * false: detection and notification is disabled.
[REMOVED] Default Value	false

3.6.1.5 WdgIf/WdgIfGeneral/WdgIfVersionInfoApi

[RTA-BSW] Description	Pre-processor switch to enable / disable the service returning the version information. true: Version information service enabled false: Version information service disabled
[AUTOSAR] Description	Pre-processor switch to enable / disable the service returning the version information.
[REMOVED] Introduction	true: Version information service enabled false: Version information service disabled

[REMOVED] Default Value	false

3.7 API Definition

List of supported API

Table 3.27.WdgIf Supported API

`WdgIf_SetMode`	Map the service WdgIf_SetMode to the service Wdg_SetMode of the corresponding Watchdog Driver.
`WdgIf_SetTriggerCondition`	Map the service WdgIf_SetTriggerCondition to the service Wdg_SetTriggerCondition of the corresponding Watchdog Driver.
`WdgIf_GetVersionInfo`	Returns the version information.

3.8 Configuration Advice

NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

3.8.1 General

- WdgIfDriverRef must reference the corresponding watchdog driver.
- Each WdgIfDevice can only be configured for a single watchdog device.

3.8.2 Non-AUTOSAR Extension

- The optional Non-AUTOSAR extension of WdgIf provides interfaces (prefixed with "WdgIf_Rb_") to access lower layer ASICs.
- Only one type of Bosch-specific ASIC can be used per project.
- The optional WdgIfRbExtWdgModuleName configuration parameter can be used to provide the name of the External Watchdog Module, which is subsequently used as the prefix for the generated APIs that will be used inside WdgIf from the Io External Device controller. If this optional parameter is not configured, no non-AUTOSAR External Watchdog interfaces will be generated.
- The WdgIfRbExtWdgApiSupport container does not require any configuration. The values for the parameters within this container will be forwarded by the External Watchdog module. If the External Watchdog module does not forward a value for these parameters, the default value of FALSE will be used

(indicating Api not supported).

- The WdgIfRbConfigSet container is optional for the Pre-compile variant.
- For any external Wdg support, as per AUTOSAR, use the `WdgIfDevice` container.

3.9 Integration Advice

WdgIf supports the generation of the WdgIf_Cfg_BSWMD.arxml file. This file can be fed to ISOLAR-AB to generate the RTE header files.

3.9.1 Init Functions

If an external watchdog is configured with a Bosch-specific ASIC under the `WdgIfGeneral` container, initialization of WdgIf is done by the ECU State Manager (EcuM).

No initialization is required for WdgIf if the project is using the AUTOSAR interfaces via the `WdgIfDevice` container.

3.9.2 DeInit Functions

No shutdown is required for WdgIf.

3.9.3 Scheduled Functions

There are no scheduled functions.

3.9.4 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

WdgIf does not use OS Resources.

3.10 Hardware Requirements

Table 3.28.WdgIf Hardware Requirements

Support	Usage
FLASH	96 bytes
RAM	0 bytes

Usage calculated based on a sample configuration.

4 Watchdog Manager

Module name	WdgM
Package	RTA-SAFE
AUTOSAR Module ID	013
AUTOSAR Version	22-11

4.1 Introduction

The Watchdog Manager (WdgM) is a basic software module (BSW) at the service layer of the standardized basic software architecture of AUTOSAR. The diagram below shows the position of WdgM in the AUTOSAR layered architecture.

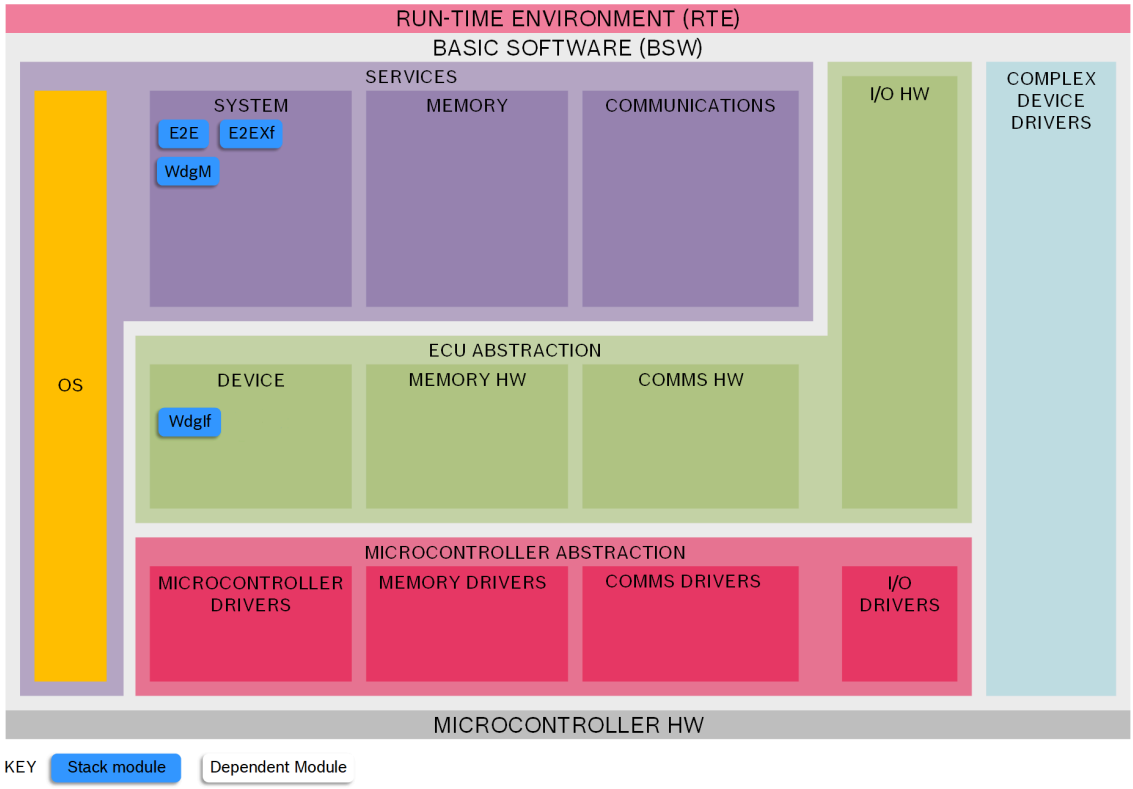

Figure 4.1.RTA-SAFE Architecture

As shown in the diagram below, WdgM is located above the Watchdog Interface (WdgIf) and the low level watchdog device drivers.

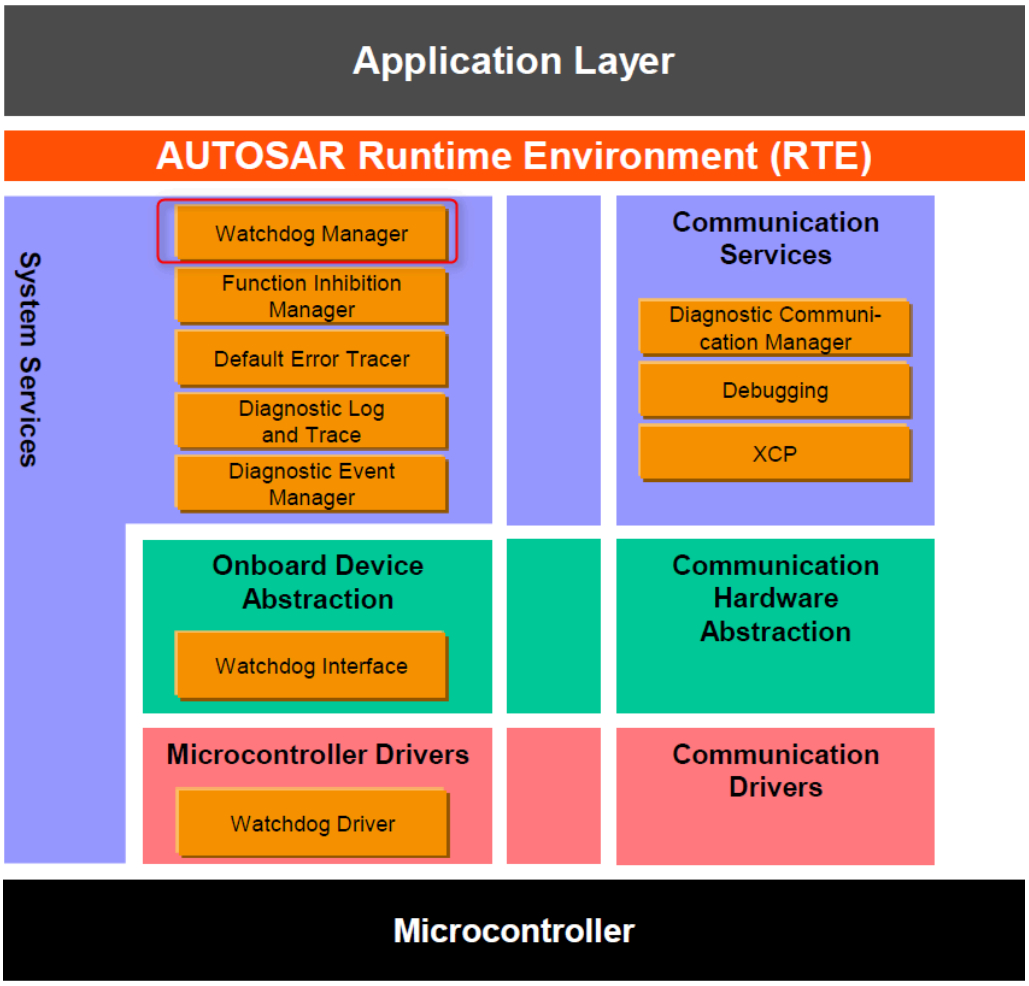

Figure 4.2.The WdgM Module

WdgM supervises the program execution of Supervised Entities (SEs) abstracting from the triggering of hardware watchdog entities. When it detects a violation of the configured temporal and/or logical constraints on program execution, it takes a number of configurable actions to recover from this failure.

Module Configuration

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

4.1.1 Interaction/Dependencies

WdgIf

WdgM uses the watchdog interface for interactions with the lower layer watchdog drivers (Wdg).

BswM

WdgM can perform a change of monitoring behaviour (via WdgM modes) in different ECU life cycles (i.e. Initial, Run, Shutdown) or user-defined modes via the

BswM. If the partition reset feature is enabled, WdgM triggers the partition reset API provided by BswM.

Mcu

When the immediate reset feature is enabled, WdgM triggers the McuReset API from the Mcu module.

If the WdgMRbDeadlineTimerSelection parameter is configured as "Mcu_Rb" then, WdgM uses the following interfaces from the Mcu for Deadline Supervision:

- Use the Mcu Macro **MCU_RB_US_TO_TIM0TICKS** to convert seconds into ticks.
- If the configuration Deadline Supervision timer interface selection is configured as an MCU, then WdgM will use the **Mcu_Rb_GetSysTicks** API to obtain the timestamp (in ticks). The assumption regarding dependent modules of WdgM like MCU is that these modules shall support Multi partition function calls and return correct value irrespective of the partition and its ASIL level these APIs are called from.
- The Frequency from the Mcu configuration ClockReferencePoint "F_MCU_SYSTEMTIMER" to convert from seconds to tick the Maximum deadline and Minimum deadline time (configured in WdgM).

OS

If WdgMRbDeadlineTimerSelection is configured as "Os", WdgM uses the GetElapsedCounterValue OS define API to perform Deadline Monitoring. The assumption regarding dependent modules of WdgM like OS is that these modules shall support Multi partition function calls and return correct value irrespective of the partition and its ASIL level these APIs are called from.

RTE

WdgM provides RTE-generated APIs for interaction with SW-C Application Software.

General

- Os - Check for critical data SEID (supervised entity ID) which is passed as a parameter to WdgM_Checkpointreached() API is done. The check for whether SEID belongs to right partition is done by comparing the partition associated with SEID with the partition X returned by OS API GetApplicationID().
- This is the assumption of use in a safety related context. Critical data that is exchanged between partitions needs to be checked if it associated with the correct partition. The user shall enable MPU as early as possible before the OS start to detect SEID associated with wrong partition. (Checkpoint reached APIs can be reported before OS start).

4.2 Supported Features

This section is divided into two separate sub-sections, describing the Autosar-defined features that are supported by WdgM, and then the additional

features provided by the RTA-BSW implementation of WdgM.

4.2.1 AUTOSAR Features

WdgM monitors both the base software and the application software. It analyzes the status of this monitoring, and it performs servicing operations of the lower layer watchdog drivers.

WdgM's supervision functionality consists of three types of supervision and monitoring -

- Alive Monitoring - To detect the frequency of the occurrence of an event.
- Deadline Monitoring - To detect the time period within which an event should be performed.
- Logical Monitoring - To monitor a program's flow path.

4.2.2 Non-AUTOSAR Features

4.2.2.1 Additional Debug Information

The RTA-BSW implementation of WdgM provides additional debug information to show which supervised entity (e.g. Core) and which checkpoint (e.g. Task Raster) is causing a failure. This additional debugging option is enabled via the WdgMRb-DebugOptionEnable configuration parameter. See the Configuration Advice section for more details.

4.2.2.2 RTE Dependency

The Mode switch indication to RTE can be optionally enabled or disabled via the WdgMRbRteDisabled configuration parameter. See the Configuration Advice section for more details.

RTE Disabled feature is supported.

4.2.2.3 Deadline Supervision Timer Interface Selection

WdgM's Deadline Supervision can use either the Autosar-defined OS interface or a proprietary Mcu interface. This is controlled by the WdgMRbDeadlineTimerSelection configuration parameter. See the Configuration Advice section for more details.

4.2.2.4 WdgM_CheckpointReached

WdgM uses OS define API GetElapsedCounterValue to perform deadline monitoring if the Deadline Supervision timer interface selection is configured as 'Os'. The return value of Api WdgM_CheckpointReached will be E_NOT_OK, if Development Error detect is enabled, then Development error will be reported with error id WDGM_E_OS_COUNTER.

4.2.2.5 Cross-Partition Access Check

- WdgM shall ensure freedom from interference between WdgM users from different EcucPartitions/ASILs.

- Master Partition Exclusive API Access:

1.	`WdgM_DeInit`
2.	`WdgM_SetMode`
3.	`WdgM_PerformReset`

- Partition-Specific Access:

1.	`WdgM_CheckpointReached`
2.	`WdgM_MainFuntion`

- Valid SEID Input Check for `WdgM_CheckpointReached` API.

4.3 Extensions

4.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

4.3.1.1 WdgM/WdgMConfigSet/WdgMDemEventParameterRefs/WDGM_E_MASTER_ALIVE_SUPERVISION

Short Name	WDGM_E_MASTER_ALIVE_SUPERVISION
Long Name	Master Alive Supervision failed detect
Description	Reference to the DemEventParameter which shall be issued when the error “Supervision has failed (Master Supervision Status has reached WDGM_MASTER_STATUS_STOPPED) and a watchdog reset will occur” has occurred.
Multiplicity	0..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

4.3.1.2 WdgM/WdgMConfigSet/WdgMMode/WdgMMainFunctionModeProps/WdgMRbMasterMainFunctionAliveSupervisionRef

Short Name	WdgMRbMasterMainFunctionAliveSupervisionRef
Long Name	Master Mainfunction Alive Supervision reference for Mode

Description	Reference to the Master Mainfunction Alive Supervision for which the WdgMMainFunctionModeProps apply.
Multiplicity	0..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

4.3.1.3 WdgM/WdgMConfigSet/WdgMMode/WdgMMainFunctionMode-Props/WdgMRbSatelliteMainFunctionAliveSupervisionRef

Short Name	WdgMRbSatelliteMainFunctionAliveSupervisionRef
Long Name	Satellite Mainfunction Alive Supervision reference for Mode
Description	Reference to the Satellite Mainfunction Alive Supervision for which the WdgMMainFunctionModeProps apply.
Multiplicity	0..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

4.3.1.4 WdgM/WdgMConfigSet/WdgMMode/WdgMSupervisionCycle

Short Name	WdgMSupervisionCycle
Long Name	WdgM MainFunction Period [OBSOLETE]
Description	[OBSOLETE] This parameter defines the schedule period of the main function WdgM_MainFunction. Unit: [s]
Multiplicity	0..1
Minimum Value	0
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

4.3.1.5 WdgM/WdgMConfigSet/WdgMMode/WdgMTrigger/WdgMTrigger-WatchDogRef

Short Name	WdgMTriggerWatchDogRef
Long Name	Watchdog reference
Description	This parameter is a reference to the configured watchdog.
Multiplicity	1..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

4.3.1.6 WdgM/WdgMConfigSet/WdgMMode/WdgMTrigger/WdgMWatchDogMode

Short Name	WdgMWatchDogMode
Long Name	Watchdog mode
Description	This parameter contains the watchdog mode that shall be used for the referenced watchdog in this Watchdog Manager mode. Implementation Type: WdgIf_ModeType
Multiplicity	1..1
Default Value	-
Enumeration Literals	WDGIF_FAST_MODE, WDGIF_OFF_MODE, WDGIF_SLOW_MODE
Type	ECUC-ENUMERATION-PARAM-DEF

4.3.1.7 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionMasterMainFunction

Short Name	WdgMRbAliveSupervisionMasterMainFunction
Long Name	Alive Supervision for Master MainFunction
Description	This container collects all configuration parameters of Alive-Supervision for Master Main function.
Multiplicity	0..65535
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

4.3.1.8 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionMasterMainFunction/WdgMRbAliveSupervisionMasterMainFunctionRef

Short Name	WdgMRbAliveSupervisionMasterMainFunctionRef
Long Name	Master Mainfunction reference for Alive Supervision
Description	Reference to the Master WdgMMainFunction for which the Alive Supervision apply.
Multiplicity	1..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

4.3.1.9 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionMasterMainFunction/WdgMRbExpiredMasterAliveCycleTol

Short Name	WdgMRbExpiredMasterAliveCycleTol

Long Name	Expired Master supervision cycle tolarance
Description	This parameter shall be used to define a value that fixes the amount of expired supervision cycles for how long the error reaction of master failure shall be postponed, After Master Supervision status has reached the state EXPIRED.
Multiplicity	1..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.10 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionMasterMainFunction/WdgMRbMasterMainFunctionExpectedAliveIndication

Short Name	WdgMRbMasterMainFunctionExpectedAliveIndication
Long Name	Expected alive indication for Master MainFunction
Description	This parameter contains the amount of expected alive indications of the Master MainFunction within the referenced amount of defined supervision cycles according to corresponding Master Mainfunction.
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.11 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionMasterMainFunction/WdgMRbMasterMainFunctionFailedAliveSupervisionRefCycleTol

Short Name	WdgMRbMasterMainFunctionFailedAliveSupervisionRefCycleTol
Long Name	Maximum failed alive supervision for Master Main Function
Description	This parameter shall contain the acceptable amount of reference cycles with incorrect/failed alive supervisions for the Master Main Function.
Multiplicity	1..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.12 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionMasterMainFunction/WdgMRbMasterMainFunctionMaxMargin

Short Name	WdgMRbMasterMainFunctionMaxMargin
Long Name	Alive indication maximum margin for Master MainFunction
Description	This parameter contains the amount of alive indications of the Master MainFunction that are acceptable to be additional to the expected alive indications within the corresponding supervision reference cycle.
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.13 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionMasterMainFunction/WdgMRbMasterMainFunctionMinMargin

Short Name	WdgMRbMasterMainFunctionMinMargin
Long Name	Alive indication minimum margin for Master MainFunction
Description	This parameter contains the amount of alive indications of the Master MainFunction that are acceptable to be missed from the expected alive indications within the corresponding supervision reference cycle.
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.14 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionMasterMainFunction/WdgMRbMasterMainFunctionSupervisionReferenceCycle

Short Name	WdgMRbMasterMainFunctionSupervisionReferenceCycle
Long Name	Supervision reference cycle for Master MainFunction
Description	This parameter shall contain the amount of supervision cycles to be used as reference by the alive-supervision mechanism to perform the checkup with counted alive indications according to corresponding Master MainFunction.
Multiplicity	1..1

Value Range	1..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.15 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionSatelliteMainFunction

Short Name	WdgMRbAliveSupervisionSatelliteMainFunction
Long Name	Alive Supervision for Satellite MainFunction
Description	This container collects all configuration parameters of Alive-Supervision for Satellite Main function. Note that each Satellite Main function may have different parameters. For example, it may have different min and max margin.
Multiplicity	0..65535
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

4.3.1.16 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionSatelliteMainFunction/WdgMRbAliveSupervisionSatelliteMainFunctionRef

Short Name	WdgMRbAliveSupervisionSatelliteMainFunctionRef
Long Name	Satellite Mainfunction reference for Alive Supervision
Description	Reference to the Satellite WdgMMainFunction for which the Alive Supervision apply.
Multiplicity	1..1
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

4.3.1.17 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionSatelliteMainFunction/WdgMRbSatelliteMainFunctionExpectedAliveIndication

Short Name	WdgMRbSatelliteMainFunctionExpectedAliveIndication
Long Name	Expected alive indication for Satellite MainFunction
Description	This parameter contains the amount of expected alive indications of the Satellite MainFunction within the referenced amount of defined supervision cycles according to corresponding Satellite Mainfunction.
Multiplicity	1..1
Value Range	0..65535

Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.18 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionSatelliteMainFunction/WdgMRbSatelliteMainFunctionFailedAliveSupervisionRefCycleTol

Short Name	WdgMRbSatelliteMainFunctionFailedAliveSupervisionRefCycleTol
Long Name	Maximum failed alive supervision for Satellite Main Function
Description	This parameter shall contain the acceptable amount of reference cycles with incorrect/failed alive supervisions for this Satellite Main Function.
Multiplicity	1..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.19 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionSatelliteMainFunction/WdgMRbSatelliteMainFunctionMaxMargin

Short Name	WdgMRbSatelliteMainFunctionMaxMargin
Long Name	Alive indication maximum margin for Satellite MainFunction
Description	This parameter contains the amount of alive indications of the Satellite MainFunction that are acceptable to be additional to the expected alive indications within the corresponding supervision reference cycle.
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.20 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionSatelliteMainFunction/WdgMRbSatelliteMainFunctionMinMargin

Short Name	WdgMRbSatelliteMainFunctionMinMargin
Long Name	Alive indication minimum margin for Satellite MainFunction
Description	This parameter contains the amount of alive indications of the Satellite MainFunction that are acceptable to be missed from the expected alive indications within the corresponding supervision reference cycle.
Multiplicity	1..1
Value Range	0..65535

Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.21 WdgM/WdgMConfigSet/WdgMRbAliveSupervisionSatelliteMainFunction/WdgMRbSatelliteMainFunctionSupervisionReferenceCycle

Short Name	WdgMRbSatelliteMainFunctionSupervisionReferenceCycle
Long Name	Supervision reference cycle for Satellite MainFunction
Description	This parameter shall contain the amount of supervision cycles to be used as reference by the alive-supervision mechanism to perform the checkup with counted alive indications according to corresponding Satellite MainFunction.
Multiplicity	1..1
Value Range	1..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.22 WdgM/WdgMGeneral/WdgMMainFunction/WdgMRbMainFunctionId

Short Name	WdgMRbMainFunctionId
Long Name	WdgM Main Function Identifier
Description	This parameter shall contain the unique identifier of the Main function.This parameter should not be configured by Integrator. Value for this parameters will be filled(forwarded) by the Id Generate Script
Multiplicity	1..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.23 WdgM/WdgMGeneral/WdgMMainFunction/WdgMRbMainFunctionType

Short Name	WdgMRbMainFunctionType
Long Name	WdgM Main Function Type

Description	This parameter distinguishes between Master and Satellite WdgMMainFunction. Note: There shall be one and only one WdgMMainFuncion configured as WDGM_MAINFUNCTION_MASTER,WDGM_MAINFUNCTION_SATELLITE_PLUS/WDGM_MAINFUNCTION_LIGHT_MASTER. WDGM_MAINFUNCTION_LIGHT_MASTER can be selected when WdgMRbLightMasterEnable is enabled else WDGM_MAINFUNCTION_SATELLITE_PLUS can be utilized.
Multiplicity	0..1
Default Value	WDGM_MAINFUNCTION_MASTER
Enumeration Literals	WDGM_MAINFUNCTION_MASTER, WDGM_MAINFUNCTION_SATELLITE_PLUS, WDGM_MAINFUNCTION_SATELLITE, WDGM_MAINFUNCTION_LIGHT_MASTER
Type	ECUC-ENUMERATION-PARAM-DEF

4.3.1.24 WdgM/WdgMGeneral/WdgMRbASILType

Short Name	WdgMRbASILType
Long Name	ASIL type selection
Description	This parameter ensures that the ASIL type is same or multiple across partition. By default same ASIL type is considered.
Multiplicity	0..1
Default Value	WDGM_RB_SAME_ASIL
Enumeration Literals	WDGM_RB_SAME_ASIL, WDGM_RB_MULTI_ASIL
Type	ECUC-ENUMERATION-PARAM-DEF

4.3.1.25 WdgM/WdgMGeneral/WdgMRbAdditionalMonitoringEnabled

Short Name	WdgMRbAdditionalMonitoringEnabled
Long Name	Additional Monitoring enable
Description	Preprocessor switch to enable/disable the WdgM Additional Monitoring feature support to external watchdogs. Shall be used to remove unneeded code segments. By default if this switch is not configured, Additional Monitoring functionality will be disabled. true: Additional Monitoring functionality is enabled. false: Additional Monitoring functionality is disabled.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.1.26 WdgM/WdgMGeneral/WdgMRbDeadlineTimeOutDetect

Short Name	WdgMRbDeadlineTimeOutDetect
Long Name	Deadline Timeout detection enable
Description	Preprocessor switch to enable/disable deadline Monitoring Timeout detection by WdgM_MainFunction, if the end checkpoint is never reached. Note: If this paremeter is set to True and if 'Os' is selected for parameter "Deadline Supervision timer interface selection" then, ensure the OsCounter referenced in all those Supervised entities containing checkpoints used in Deadline Supervision, are accessible by the Os Application in whose context WdgM_MainFunction() is called Bugzilla reference: http://www.autosar.org/bugzilla/show_bug.cgi?id=56735 This is an Parameter.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.1.27 WdgM/WdgMGeneral/WdgMRbDeadlineTimerSelection

Short Name	WdgMRbDeadlineTimerSelection
Long Name	Deadline Supervision timer interface selection

Description	Selection of the Base software module whose interface to be used, to access timer for Deadline supervision. This parameter is not required to be configured if Deadline supervision < WdgMDeadlineSupervision > is not configured. Mcu_Rb: Timer is accessed via vendor (RTA-BSW) specific Mcu Interface. Non Autosar solution. Reduces interrupt/multicore locking duration when WdgM_CheckPointReached() Api is called. For DGS it is recommended to select this option. Note: If this option is choosen , WdgM Os Counter reference < WdgMOSCounter > should not be configured in any Supervised entity < WdgMSupervisedEntity >. Os: Timer is accessed via Os interface. Autosar solution. Note: If this option is choosen, WdgM Os Counter reference < WdgMOSCounter > should be configured in each Supervised entity < WdgMSupervisedEntity > Whose checkpoint/s is/are part of Deadline supervision < WdgMDeadlineSupervision >. If this parameter is not configured then option “Os” is assumed.
Multiplicity	0..1
Default Value	-
Enumeration Literals	MCU_RB_TIMER, AUTOSAR_OS_TIMER
Type	ECUC-ENUMERATION-PARAM-DEF

4.3.1.28 WdgM/WdgMGeneral/WdgMRbDebugOptionEnable

Short Name	WdgMRbDebugOptionEnable
Long Name	Debug Option Enable
Description	Preprocessor switch to enable/disable the Debug option. true: Debug Option is enabled. false: Debug Option is disabled.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.1.29 WdgM/WdgMGeneral/WdgMRbInterruptLockDisable

Short Name	WdgMRbInterruptLockDisable
Long Name	Interrupt Lock disable for WdgM_MainFunction [OBSOLETE]

Description	[OBSOLETE] Preprocessor switch to Enable/Disable the Interrupt Lock with WdgM MainFunction. True: In your use case WdgM_Main function cannot be preempted by WdgM_SetMode vice versa, if WdgM_SetMode is not preempted by Main-Function then keep this switch true i.e Interrupt Lock Disabled. False: Interrupt lock is required in WdgM_MainFunction since SetMode can preempt it vice versa.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.1.30 WdgM/WdgMGeneral/WdgMRbLightMasterEnable

Short Name	WdgMRbLightMasterEnable
Long Name	Light Master Enable
Description	Preprocessor switch to enable/disable the LightMaster feature. When enabled WatchDogs trigger are stopped via Light Master instead of master. true: LightMaster is enabled. false: LightMaster is disabled
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.1.31 WdgM/WdgMGeneral/WdgMRbLogicalDynamicBehavior

Short Name	WdgMRbLogicalDynamicBehavior
Long Name	Logical Supervision dynamic behavior enable
Description	Preprocessor switch to enable/disable the Logical Supervision dynamic behavior scenario during mode switch. true: Logical Supervision dynamic behavior is enabled. false: Logical Supervision dynamic behavior is disabled.
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.1.32 WdgM/WdgMGeneral/WdgMRbMasterDemStoppedSupervision-Report

Short Name	WdgMRbMasterDemStoppedSupervisionReport
Long Name	DEM supervision support for master alive supervision

Description	Parameter to enable/disable the error reporting to DEM for master alive supervision. true: A notification to DEM is sent if the Master Watchdog Manager reaches the state WDGM_MASTER_STATUS_STOPPED. false: The notification is disabled.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.1.33 WdgM/WdgMGeneral/WdgMRbMasterImmediateReset

Short Name	WdgMRbMasterImmediateReset
Long Name	Immediate reset enable for master alive supervision
Description	This parameter enables/disables the immediate reset feature in case of master alive supervision failure. true: Immediate reset for master alive supervision is enabled false: Immediate reset for master alive supervision is disabled. Additional information: If this parameter is set to TRUE and Master Supervision Status reaches the state WDGM_MASTER_STATUS_STOPPED, the Watchdog Manager module will call the MCU service Mcu_PerformReset. This parameter can be set to TRUE only if parameter McuPerformResetApi is set to TRUE (McuPerformResetApi is set to TRUE implies, service Mcu_PerformReset will be available).
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.1.34 WdgM/WdgMGeneral/WdgMRbOsTask

Short Name	WdgMRbOsTask
Long Name	Supervised entity Os Task
Description	This container collects all Os Tasks under which the Supervised Entities are mapped to.
Multiplicity	0..65535
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

4.3.1.35 WdgM/WdgMGeneral/WdgMRbOsTask/WdgMRbOsTaskMaxCPReportPerMainFuncCycle

Short Name	WdgMRbOsTaskMaxCPReportPerMainFuncCycle
Long Name	Maximum number of Checkpoints within one cycle of main function
Description	Maximum number of checkpoints expected to be reported from the OsTask within one cycle of corresponding WdgM MainFunction (considering all configured WdgM Modes). This parameter shall be configured by integrator. If not default value shall be set by WdgM.
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.36 WdgM/WdgMGeneral/WdgMRbOsTask/WdgMRbOsTaskMaxCPReportPerOsTaskCycle

Short Name	WdgMRbOsTaskMaxCPReportPerOsTaskCycle
Long Name	Maximum number of Checkpointreached per one execution of Os task
Description	This parameter contains the maximum number of Checkpointreached per 1 execution Os task. This parameter shall be configured by integrator. If not default value shall be set by WdgM.
Multiplicity	0..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.37 WdgM/WdgMGeneral/WdgMRbOsTask/WdgMRbOsTaskPeriod

Short Name	WdgMRbOsTaskPeriod
Long Name	Task period of the monitored Os task
Description	This parameter specifies the period in seconds of the task which is monitored by the supervised entity and is mandatory in case of single ASIL use case of WdgM.
Multiplicity	0..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

4.3.1.38 WdgM/WdgMGeneral/WdgMRbOsTask/WdgMRbOsTaskRef

Short Name	WdgMRbOsTaskRef
Long Name	Supervised entity Os Task reference
Description	Denotes in which "OsTask" the supervised entity is executed. This parameter is mandatory in case of Multi ASIL use case of WdgM.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

4.3.1.39 WdgM/WdgMGeneral/WdgMRbRteDisabled

Short Name	WdgMRbRteDisabled
Long Name	RTE disable
Description	Preprocessor switch to enable/disable the Mode switch indication to RTE. Shall be used to remove unneeded code segments. By default if this switch is not configured, RTE mode indication will be enabled. true: Mode switch indication is disabled false: Mode switch indication is enabled. Note: If configuration parameter "EcuCRbRteInUse" is provided in container "EcuCRbSettings" of module "EcuC" then, Mode switch Indication can be enabled only if "EcuCRbRteInUse" is set to True.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

4.3.1.40 WdgM/WdgMGeneral/WdgMRbRtePortInterfaces

Short Name	WdgMRbRtePortInterfaces
Long Name	Rte Port Interfaces
Description	This container contains the Queue Length parameters for port instances that are created for different operations and modes.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

4.3.1.41 WdgM/WdgMGeneral/WdgMRbRtePortInterfaces/WdgMRbCheckpointReachedQueueLength

Short Name	WdgMRbCheckpointReachedQueueLength
Long Name	CheckpointReached Queue Length
Description	This parameter shall contain the size of the queue for each port instance that is created for CheckpointReached operation. If the parameter is not configured by the integrator then default value 1 shall be assumed.
Multiplicity	0..1
Value Range	1..65535
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.42 WdgM/WdgMGeneral/WdgMRbRtePortInterfaces/WdgMRbGetFirstExpiredMainFunctIDQueueLength

Short Name	WdgMRbGetFirstExpiredMainFunctIDQueueLength
Long Name	GetFirstExpiredMainFunctID Queue Length
Description	This parameter shall contain the size of the queue for each port instance that is created for GetFirstExpired MainFunctID operation. If the parameter is not configured by the integrator then default value 1 shall be assumed.
Multiplicity	0..1
Value Range	1..65535
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.43 WdgM/WdgMGeneral/WdgMRbRtePortInterfaces/WdgMRbGetFirstExpiredSEIDQueueLength

Short Name	WdgMRbGetFirstExpiredSEIDQueueLength
Long Name	GetFirstExpiredSEID Queue Length
Description	This parameter shall contain the size of the queue for each port instance that is created for GetFirstExpiredSEID operation. If the parameter is not configured by the integrator then default value 1 shall be assumed.
Multiplicity	0..1
Value Range	1..65535

Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.44 WdgM/WdgMGeneral/WdgMRbRtePortInterfaces/WdgMRbGetGlobalStatusQueueLength

Short Name	WdgMRbGetGlobalStatusQueueLength
Long Name	GetGlobalStatus Queue Length
Description	This parameter shall contain the size of the queue for each port instance that is created for GetGlobalStatus operation. If the parameter is not configured by the integrator then default value 1 shall be assumed.
Multiplicity	0..1
Value Range	1..65535
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.45 WdgM/WdgMGeneral/WdgMRbRtePortInterfaces/WdgMRbGetLocalStatusQueueLength

Short Name	WdgMRbGetLocalStatusQueueLength
Long Name	GetLocalStatus Queue Length
Description	This parameter shall contain the size of the queue for each port instance that is created for GetLocalStatus operation. If the parameter is not configured by the integrator then default value 1 shall be assumed.
Multiplicity	0..1
Value Range	1..65535
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.46 WdgM/WdgMGeneral/WdgMRbRtePortInterfaces/WdgMRbGetModeQueueLength

Short Name	WdgMRbGetModeQueueLength
Long Name	GetMode Queue Length
Description	This parameter shall contain the size of the queue for each port instance that is created for GetMode operation. If the parameter is not configured by the integrator then default value 1 shall be assumed.

Multiplicity	0..1
Value Range	1..65535
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.47 WdgM/WdgMGeneral/WdgMRbRtePortInterfaces/WdgMRbModeSwitchPortGlobalModeQueueLength

Short Name	WdgMRbModeSwitchPortGlobalModeQueueLength
Long Name	WdgM_GlobalMode Mode Switch Port Queue Length
Description	This parameter shall contain the size of the queue for the globalMode port of WdgM_GlobalMode mode switch port interface. If the parameter is not configured by the integrator then default value 1 shall be assumed.
Multiplicity	0..1
Value Range	1..65535
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.48 WdgM/WdgMGeneral/WdgMRbRtePortInterfaces/WdgMRbModeSwitchPortLocalModeQueueLength

Short Name	WdgMRbModeSwitchPortLocalModeQueueLength
Long Name	WdgM_LocalMode Mode Switch Port Queue Length
Description	This parameter shall contain the size of the queue for each port instance that is created under WdgM_LocalMode mode switch port interface. If the parameter is not configured by the integrator then default value 1 shall be assumed.
Multiplicity	0..1
Value Range	1..65535
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.49 WdgM/WdgMGeneral/WdgMRbRtePortInterfaces/WdgMRbPerformResetQueueLength

Short Name	WdgMRbPerformResetQueueLength
Long Name	PerformReset Queue Length

Description	This parameter shall contain the size of the queue for each port instance that is created for PerformReset operation. If the parameter is not configured by the integrator then default value 1 shall be assumed.
Multiplicity	0..1
Value Range	1..65535
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.50 WdgM/WdgMGeneral/WdgMRbRtePortInterfaces/WdgMRbSetModeQueueLength

Short Name	WdgMRbSetModeQueueLength
Long Name	SetMode Queue Length
Description	This parameter shall contain the size of the queue for each port instance that is created for SetMode operation. If the parameter is not configured by the integrator then default value 1 shall be assumed.
Multiplicity	0..1
Value Range	1..65535
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

4.3.1.51 WdgM/WdgMGeneral/WdgMSupervisedEntity/WdgMEcucPartitionRef

Short Name	WdgMEcucPartitionRef
Long Name	Supervised entity partition reference [OBSOLETE]
Description	[OBSOLETE] Denotes in which “EcucPartition” the supervised entity is executed. When the partition is stopped, the supervised entity shall be de-activated in the WdgM to avoid an ECU reset.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

4.3.1.52 WdgM/WdgMGeneral/WdgMSupervisedEntity/WdgMInternalCheckpointFinalRef

Short Name	WdgMInternalCheckpointFinalRef
Long Name	Final checkpoint
Description	This is the reference to the final Checkpoint(s) for this Supervised Entity.

Multiplicity	0..65535
Default Value	-
Type	ECUC-SYMBOLIC-NAME-REFERENCE-DEF

4.3.1.53 WdgM/WdgMGeneral/WdgMSupervisedEntity/WdgMOsApplicationRef

Short Name	WdgMOsApplicationRef
Long Name	OS application reference [OBSOLETE]
Description	[OBSOLETE] Optional reference to an OS Application. Beware, the Watchdog Manager module will trigger a partition restart of this OS Application when the corresponding Supervised Entity reaches WDGM_LOCAL_STATUS_FAILED.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

4.3.1.54 WdgM/WdgMGeneral/WdgMSupervisedEntity/WdgMRbMonitoredOsTaskRef

Short Name	WdgMRbMonitoredOsTaskRef
Long Name	Supervised entity OsTask reference
Description	Denotes under which “Ostask” the supervised entity is executed.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

4.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

4.3.2.1 WdgM_RB_GetMasterStatus

Syntax	`WdgM_RB_GetMasterStatus`
Parameters	(OUT) MasterStatus
Return Value	NONE
Description	Returns the master supervision status of master main function.

4.3.2.2 WdgM_RB_GetFirstExpiredMainFunctID

Syntax	`WdgM_RB_GetFirstExpiredMainFunctID`

Parameters	(OUT) MainfunctionID
Return Value	NONE
Description	Returns MainfunctionID that first reached the state WDGM_LOCAL_STATUS_EXPIRED.

4.4 Limitations

The following features are not supported for version v12.7.0:

4.4.1 General

Post-build loadable feature is not supported and is permanent deviation from AUTOSAR.

16-bit Micro controller architecture is not supported.

WdgM is not following a 'one fits all' approach i.e. this implementation is inline with AR v22-11 and integration to another version is not allowed.

In case of multiple WdgM_MainFunction instances, Satellite WdgM_MainFunction instances can be called at any time, regardless of the concurrent invocation of the function WdgM_SetMode in the partition corresponding to Master WdgM_MainFunction instance.

Re-Initialisation (even after call of WdgM_DeInit) is not supported.

One WdgM_MainFunction is allowed per EcucPartition.

WdgM_SetMode callable only in the same Task as Master WdgM_MainFunction.

On mode change, Master Supervision status transition from EXPIRED to STOPPED duration may vary.

Assumptions

1. Master WdgM_MainFunction runs in the highest ASIL Partition context.
2. No Cross Partition Logical Supervision configuration.
3. An EcucPartition is confined to one Core (not spread across cores).

Development Error Detection: To prevent FFI or MPU traps when the MPU is activated, it is recommended to disable the WdgM Development Error (`WdgMDevErrorDetect`) as the DET API used in WdgM is designed to be called only from the Master Partition.

4.4.2 Alive Supervision

Alive supervision is inactive during mode change. It is disabled during the period between a mode change request and the next scheduled WdgM_MainFunction call.

4.4.3 Alive Overflow Protection

During the action of alive monitoring analysis for a checkpoint (i.e. WdgM_MainFunction), if the occurrence of alive indication is greater than uint32, no roll-over protection is available for alive indication.

4.4.4 Deadline Monitoring

When configuring Deadline Monitoring, the value of WdgMDeadlineMax is limited by the roll-over value of the counter used in the GetElapsedCounterValue API.

Deadline monitoring TimeOut detection is possible for only those supervised entities where the OsCounter is accessible from WdgM_MainFunction.

If Deadline monitoring is configured and value 'Mcu_Rb' is selected for configuration parameter "Deadline Supervision timer interface selection(WdgMRbDeadlineTimerSelection)", then note that, WdgM does not support variation in the value of clock frequency of "F_STM", "F_TIMEBASE" and "F_MCU_SYSTIMER" (for IFX, ST and MDG1 controllers respectively) across post build variant configuration of Mcu.

4.4.5 Deadline Supervision

The start and stop checkpoint should not be configured as per deadline supervision. This is because the deadline supervision can give false positive results if the mainfunction is interrupted by a high priority task.

4.4.6 Internal Logical Supervision

All checkpoints belonging to an Internal graph should be reported from the same Task. The concurrent reporting of checkpoints of an Internal graph is not supported.

All checkpoints of an Internal Graph (Logical Supervision) should belong to same FIFO to which the supervised entity is mapped to.

The validation check for Internal graph cannot currently detect breaks in the graph when loops have been configured, so you should ensure that no breaks are configured in the Internal graph.

A Checkpoint must not belong to more than one Internal Graph.

4.4.7 Internal Logical Supervision Safety Risk

WdgM Internal Logical Supervision works successfully after every Mode switch irrespective of what checkpoint is reported first (Initial/Intermediate/Final Checkpoint). Because of this, it cannot monitor the missed checkpoint (due to trap/hang) in the graph. Therefore a skipped checkpoint may be a critical checkpoint.

4.4.8 External Logical Supervision

All checkpoints belonging to an External graph should be reported from the same Task. The concurrent reporting of checkpoints of an External graph is not supported.

The validation check for External graph cannot currently detect breaks in the graph when loops have been configured, so you should ensure that no breaks are configured in the External graph.

4.4.9 External Logical Supervision Safety Risk

WdgM External Logical Supervision works successfully after every Mode switch irrespective of what checkpoint is reported first (Initial/Intermediate/Final Checkpoint). Because of this, it cannot monitor the missed checkpoint (due to trap/hang) in the graph. Therefore a skipped checkpoint may be a critical checkpoint.

4.4.10 WdgM_DeInit

WdgM_DeInit does not update the trigger condition of Watchdog Drivers.

4.4.11 WdgM_CheckPointReached

To avoid overly long locking periods (i.e. > 2us) the same checkpoint should not be part of multiple Deadline Supervisions.

4.5 Exclusions

The following features from AR v22-11 are not supported in version v12.7.0.

4.5.1 Configuration Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- WdgM/WdgMConfigSet/WdgMMode/WdgMAliveSupervision/WdgMMaxMargin
- WdgM/WdgMConfigSet/WdgMMode/WdgMAliveSupervision/WdgMSupervisionReferenceCycle
- WdgM/WdgMConfigSet/WdgMMode/WdgMDeadlineSupervision/WdgMDeadlineMax
- WdgM/WdgMConfigSet/WdgMMode/WdgMDeadlineSupervision/WdgMDeadlineMin
- WdgM/WdgMConfigSet/WdgMMode/WdgMExpiredSupervisionCycleTol
- WdgM/WdgMConfigSet/WdgMMode/WdgMExternalLogicalSupervision/WdgMCrossClusterTransition
- WdgM/WdgMConfigSet/WdgMMode/WdgMExternalLogicalSupervision/WdgMCrossClusterTransition/WdgMCrossClusterTransitionDestRef
- WdgM/WdgMConfigSet/WdgMMode/WdgMExternalLogicalSupervision/WdgMCrossClusterTransition/WdgMCrossClusterTransitionSourceRef
- WdgM/WdgMConfigSet/WdgMMode/WdgMExternalLogicalSupervision/WdgMExternalCheckpointInitialRef
- WdgM/WdgMConfigSet/WdgMMode/WdgMExternalLogicalSupervision/WdgMTransitionProxy
- WdgM/WdgMConfigSet/WdgMMode/WdgMExternalLogicalSupervision/WdgMTransitionProxy/WdgMResourceRef

- WdgM/WdgMConfigSet/WdgMMode/WdgMLocalStatusParams/WdgMFailedAliveSupervisionRefCycleTol
- WdgM/WdgMConfigSet/WdgMMode/WdgMTrigger/WdgMTriggerWatchdogRef
- WdgM/WdgMConfigSet/WdgMMode/WdgMTrigger/WdgMWatchdogMode
- WdgM/WdgMGeneral/WdgMBaseSocket
- WdgM/WdgMGeneral/WdgMBaseSocket/WdgMResourceRef
- WdgM/WdgMGeneral/WdgMBaseSocket/WdgMSocketEcucPartitionRef
- WdgM/WdgMGeneral/WdgMEnableTimeoutDetection
- WdgM/WdgMGeneral/WdgMSupervisedEntity/WdgMInternalCheckpointFinalRef
- WdgM/WdgMGeneral/WdgMSwClusterSupport

4.6 Deviations

4.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

4.6.1.1 WdgM

[RTA-BSW] Description	Configuration of the WdgM (Watchdog Manager) module (with vendor specific enhancements of RB).
[AUTOSAR] Description	Configuration of the WdgM (Watchdog Manager) module.

4.6.1.2 WdgM/WdgMConfigSet/WdgMDemEventParameterRefs

[RTA-BSW] Description	Container for the references to DemEventParameter elements which shall be invoked using the API Dem_SetEventStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.
[AUTOSAR] Description	Container for the references to DemEventParameter elements which shall be invoked using the API Dem_SetEventStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId symbolic value. The standardized errors are provided in this container and can be extended by vendor-specific error references.

4.6.1.3 WdgM/WdgMConfigSet/WdgMMode/WdgMAliveSupervision/WdgMMinMargin

[RTA-BSW] Description	This parameter contains the amount of alive indications of the Checkpoint that are acceptable to be missed from the expected alive indications within the corresponding supervision reference cycle. Range specified by Autosar: 0 - 255 It is extended to: 0 - 65535.
[AUTOSAR] Description	This parameter contains the amount of alive indications of the Checkpoint that are acceptable to be missed from the expected alive indications within the corresponding supervision reference cycle.
[RTA-BSW] Maximum Value	65535
[AUTOSAR] Maximum Value	255

4.6.1.4 WdgM/WdgMConfigSet/WdgMMode/WdgMExternalLogicalSupervision

[RTA-BSW] Description	This container collects all configuration parameters for Logical Supervision for one external graph
[AUTOSAR] Description	This container collects all configuration parameters for Logical Supervision for one external graph.

4.6.1.5 WdgM/WdgMConfigSet/WdgMMode/WdgMExternalLogicalSupervision/WdgMExternalCheckpointFinalRef

[RTA-BSW] Description	This is the reference to the final Checkpoint(s) for this External Graph.
[AUTOSAR] Description	This is the reference to the final Checkpoint(s) for this External Graph which can end with a WdgMCheckpoint or in case of cross cluster transitions with a WdgMTransitionProxy.
[REMOVED] Introduction	Both WdgMCheckpoint(s) and WdgMTransitionProxy(s) could be mixed inside the same WdgMExternalLogicalSupervision.

4.6.1.6 WdgM/WdgMConfigSet/WdgMMode/WdgMMainFunctionModeProps

[RTA-BSW] Description	This container provides configuration values for a WdgMMainFunction which apply in a specific WdgMMode.

[AUTOSAR] Description	This container provides configuration values for a WdgMMainFunction which apply in a specific WdgMMode. Tags: atp.Status=draft
[RTA-BSW] Multiplicity	1..255
[AUTOSAR] Multiplicity	0..*

4.6.1.7 WdgM/WdgMConfigSet/WdgMMode/WdgMMainFunctionModeProps/WdgMMainFunctionModePropsMainFunctionRef

[RTA-BSW] Description	Reference to the WdgMMainFunction for which the WdgMMainFunctionModeProps apply.The reference to WdgMMainfunction container is forwarded to WdgMMainFunctionModePropsMainFunctionRef container by the script in case of same ASIL configuration, thus it need not be configured for a same ASIL use case.
[AUTOSAR] Description	Reference to the WdgMMainFunction for which the WdgMMainFunctionModeProps apply. Tags: atp.Status=draft
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

4.6.1.8 WdgM/WdgMConfigSet/WdgMMode/WdgMMainFunctionModeProps/WdgMMainFunctionModePropsTimePeriod

[RTA-BSW] Description	The period between successive calls to according instance of WdgM_MainFunction in seconds.
[AUTOSAR] Description	The period between successive calls to according instance of WdgM_MainFunction in seconds. This parameter may be used by the WdgM generator to transform the values of the WdgMModes and/or WdhMSupervisedEntities timing configuration parameters of the WdgM module to internal implementation specific counter or tick values. The WdgM module's internal timing handling is implementation specific. The WdgM module (generator) may rely on the fact that Wdg_MainFunction is scheduled according to the value configured here. Tags: atp.Status=draft
[REMOVED] Introduction	Unit: [s]
[REMOVED] Maximum Value	INF

4.6.1.9 WdgM/WdgMConfigSet/WdgMMode/WdgMModeId

[RTA-BSW] Description	This parameter fixes the identfier for the mode. This identifier is for instance passed as a parameter to the WdgM_SetMode service.This parameter should not be configured by Integrator. Value for this parameters will be filled(forwarded) by the Id Generate Script
[AUTOSAR] Description	This parameter fixes the identifier for the mode. This identifier is for instance passed as a parameter to the WdgM_SetMode service.

4.6.1.10 WdgM/WdgMConfigSet/WdgMMode/WdgMTrigger/WdgMTrigger-ConditionValue

[RTA-BSW] Description	This parameter shall contain the value that is passed to WdgIf_SetTriggerCondition for this watchdog. Unit [msec]
[AUTOSAR] Description	This parameter shall contain the value that is passed to WdgIf_SetTriggerCondition for this watchdog.

4.6.1.11 WdgM/WdgMGeneral/WdgMDevErrorDetect

[RTA-BSW] Description	Switches the development error detection and notification on or off. true: detection and notification is enabled. false: detection and notification is disabled.
[AUTOSAR] Description	Switches the development error detection and notification on or off.
[REMOVED] Introduction	* true: detection and notification is enabled. * false: detection and notification is disabled.
[REMOVED] Default Value	false

4.6.1.12 WdgM/WdgMGeneral/WdgMImmediateReset

[RTA-BSW] Description	This parameter enables/disables the immediate reset feature in case of alive-supervision failure. true: Immediate reset is enabled false: Immediate reset is disabled. Additional information: If this parameter is set to TRUE and Global Supervision Status reaches the state WDGM_GLOBAL_STATUS_STOPPED, the Watchdog Manager module will call the MCU service Mcu_PerformReset. This parameter can be set to TRUE only if parameter McuPerformResetApi is set to TRUE (McuPerformResetApi is set to TRUE implies, service Mcu_PerformReset will be available).

[AUTOSAR] Description	This parameter enables/disablse the immediate reset feature in case of alive-supervision failure.
[REMOVED] Introduction	true: Immediate reset is enabled false: Immediate reset is disabled
[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0

4.6.1.13 WdgM/WdgMGeneral/WdgMMainFunction

[RTA-BSW] Description	Reference to the WdgMInstanceMainFunction which this Supervised Entity belongs to. Relevant to Alive Supervision and Deadline Supervision.In case of same asil a mainfunction container is forwarded.
[AUTOSAR] Description	Reference to the WdgMInstanceMainFunction which this Supervised Entity belongs to. Relevant to Alive Supervision and Deadline Supervision Tags: atp.Status=draft

4.6.1.14 WdgM/WdgMGeneral/WdgMMainFunction/WdgMMainFunction-PartitionRef

[RTA-BSW] Description	Reference to EcucPartition, where the according WdgM_MainFunction instance is assigned to. For the software architecture with single partition, this reference is unnecessary.
[AUTOSAR] Description	Reference to EcucPartition, where the according WdgM_MainFunction instance is assigned to. For the software architecture with single partition, this reference is unnecessary. Tags: atp.Status=draft

4.6.1.15 WdgM/WdgMGeneral/WdgMOffModeEnabled

[RTA-BSW] Description	This parameter enables/disables the selection of the “OffMode” of the watchdog driver. true: “OffMode” selection is allowed false: “OffMode” selection is disallowed.
[AUTOSAR] Description	This parameter enables/disables the selection of the “OffMode” of the watchdog driver.
[REMOVED] Introduction	true: “OffMode” selection is allowed false: “OffMode” selection is disallowed
[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0

4.6.1.16 WdgM/WdgMGeneral/WdgMSupervisedEntity/WdgMCheckpoint/WdgMCheckpointId

[RTA-BSW] Description	This parameter shall contain the unique identifier of Checkpoint.This parameter should not be configured by Integrator. Value for this parameters will be filled(forwarded) by the Id Generate Script
[AUTOSAR] Description	This parameter shall contain the unique identifier of Checkpoint.

4.6.1.17 WdgM/WdgMGeneral/WdgMSupervisedEntity/WdgMMainFunctionRef

[RTA-BSW] Description	This is the reference to the WdgMInstanceMainFunction which this Supervised Entity belongs to. Relevant to Alive Supervision and Deadline Supervision. The referece to WdgMMainfunction is forwarded to WdgMMainFunctionRef containers in case of same ASIL configuration
[AUTOSAR] Description	Reference to the WdgMInstanceMainFunction which this Supervised Entity belongs to. Relevant to Alive Supervision and Deadline Supervision Tags: atp.Status=draft

4.6.1.18 WdgM/WdgMGeneral/WdgMSupervisedEntity/WdgMOSCounter

[RTA-BSW] Description: OS counter used by Watchdog Manager to perform the deadline monitoring of the Supervised Entity. Note: 1) Do not configure this parameter If value selected for configuration parameter “Deadline Supervision timer interface selection” is “Mcu_Rb”. 2) It’s mandatory to configure this parameter if, value selected for configuration parameter “Deadline Supervision timer interface selection” is “Os” and “Deadline Supervision” is configured using checkpoint/s of this Supervised Entity. In the above mentioned case, also ensure the following. a) Os Application in whose context WdgM_CheckPointReached() is called for the checkpoint/s (which is/are part of Deadline Supervision) of this Supervised entity, either owns or needs to be configured (OsCounterAccessingApplication) to access this OsCounter. b) If parameter “Deadline Timeout detection enable” is set to True then, ensure that the Os Application in whose context WdgM_MainFunction() is called, either owns or has access to this OsCounter

[AUTOSAR] Description: OS counter used by Watchdog Manager to perform the deadline supervision of the Supervised Entity.

4.6.1.19 WdgM/WdgMGeneral/WdgMSupervisedEntity/WdgMSupervisedEntityId

[RTA-BSW] Description	This parameter shall contain the unique identifier of the supervised entity.This parameter should not be configured by Integrator. Value for this parameters will be filled(forwarded) by the Id Generate Script
[AUTOSAR] Description	This parameter shall contain the unique identifier of the supervised entity.

4.6.1.20 WdgM/WdgMGeneral/WdgMVersionInfoApi

[RTA-BSW] Description	Preprocessor switch to enable/disable the existence of the API WdgM_GetVersionInfo. Shall be used to remove unneeded code segments. true: API is enabled false: API is disabled
[AUTOSAR] Description	Preprocessor switch to enable/disable the existence of the API WdgM_GetVersionInfo. Shall be used to remove unneeded code segments.
[REMOVED] Introduction	true: API is enabled false: API is disabled
[REMOVED] Default Value	false

4.6.1.21 WdgM/WdgMGeneral/WdgMWatchdog/WdgMWatchdogName

[RTA-BSW] Description	This parameter shall contain the symbolic name of the watchdog instance.This parameter should not be configured by Integrator. Value for this parameters will be filled(forwarded) by the Id Generate Script
[AUTOSAR] Description	This parameter shall contain the name of the watchdog instance.

4.7 API Definition

Table 4.80.WdgM Supported API functions

`WdgM_Init`	Initializes the module.
`WdgM_DeInit`	De-initializes the module.
`WdgM_GetVersionInfo`	Returns the version information of this module.
`WdgM_SetMode`	Sets the current mode of the Watchdog Manager.
`WdgM_GetMode`	Returns the current mode of the Watchdog Manager.
`WdgM_CheckpointReached`	Indicates to the Watchdog Manager that a Checkpoint within a Supervised Entity has been reached.
`WdgM_GetLocalStatus`	Returns the supervision status of an individual Supervised Entity.
`WdgM_GetGlobalStatus`	Returns the global supervision status of the Watchdog Manager.
`WdgM_PerformReset`	Instructs the Watchdog Manager to cause a watchdog reset.
`WdgM_GetFirstExpiredSEID`	Returns the SEID that first reached the state WDGM_LOCAL_STATUS_EXPIRED.
`WdgM_MainFunction`	Performs the processing of the cyclic Watchdog Manager jobs.

For further information on these functions, please refer to AUTOSAR_SWS_WatchdogManager.pdf

4.8 Configuration Advice

> **NOTE**
>
> Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

4.8.1 General

- At least one Supervised Entity (WdgMSupervisedEntity) should be configured.
- WdgMSupervisionCycle must always be configured using a float value.

4.8.2 WdgMMaxMargin and WdgMMinMargin

Although the Autosar-defined range of WdgMMaxMargin and WdgMMinMargin is from 0 to 255, the range supported in RTA-BSW's implementation of WdgM is from 0 to 65535.

4.8.3 WdgM_SupervisedEntityIdType

The mapping of the WdgM_SupervisedEntityIdType data type is based on the number of Supervised Entities that have been configured.

- If there are more than 255 Supervised Entities, WdgM_SupervisedEntityIdType is mapped to uint16 (as defined in Autosar).
- If there are less than or equal to 255 Supervised Entities, WdgM_SupervisedEntityIdType is mapped to unit8 to reduce the amount of memory needed to store the Supervised Entity IDs.

4.8.4 WdgM_CheckpointIdType

The mapping of the WdgM_CheckpointIdType data type is based on the number of Checkpoints that have been configured.

- If there are more than 255 Checkpoints, WdgM_CheckpointIdType is mapped to unint16 (as defined in Autosar).
- If there are less than or equal to 255 Checkpoints, WdgM_CheckpointIdType is mapped to unit8 to reduce the amount of memory needed to store the Checkpoint IDs.

4.8.5 WdgMRbInterruptLockDisable

WdgM MainFunction has two modes of operation, as controlled by the WdgM-RbInterruptLockDisable configuration parameter.

- WdgMRbInterruptLockDisable should be set to TRUE if WdgM_MainFunction and WdgM_SetMode neither operate in parallel nor pre-empt each other. The WdgM_MainFunction locking period is small and does not heavily impact the system.
- WdgMRbInterruptLockDisable should be set to FALSE if WdgM_MainFunction and WdgM_SetMode can operate in parallel or can pre-empt each other. In this case, WdgM_MainFunction's locking period will be approximately 30usec.

4.8.6 WdgMRbDeadlineTimerSelection

The WdgMRbDeadlineTimerSelection configuration parameter allows for the selection of the Deadline Supervision timer interface.

- If WdgMRbDeadlineTimerSelection is configured as "Mcu_Rb", WdgM's Deadline Supervision uses a proprietary Mcu Interface (instead of the Autosar-defined Os interface) to obtain the timestamp.
- If WdgMRbDeadlineTimerSelection is configured as "Os", WdgM uses the Autosar-defined API GetElapsedCounterValue to perform the deadline monitoring. If GetElapsedCounterValue returns anything other than E_OK, the storing of the time stamp and the evaluation of Deadline Supervision is not done. The return value of WdgM_CheckpointReached will be E_NOT_OK. If Development Error detection is enabled, development errors will be reported with the error id "WDGM_E_OS_COUNTER".
- Care should be taken when configuring deadline maximum values based on Mcu clock configurations (resolution and overflow).

4.8.7 WdgMRbDebugOptionEnable

RTA-BSW's implementation of WdgM provides additional debug information to show which supervised entity (e.g. Core) and which checkpoint (e.g. Task Raster) is causing a failure. This additional debugging option is enabled via the WdgMRb-DebugOptionEnable configuration parameter.

The "WdgM_Rb_FirstExpiredEntityInfo_st" structure is used to provide the debug information (see "WdgM_Rb_FirstExpiredEntityInfo_tst" for type information).

The tables below provide detailed information for this additional debugging:

Name of Element of Debug Structure	Description
FirstExpiredLocation_t	Indicates Type of Supervision failure.
FirstExpiredSupervisedEntityId_t	ID of the first expired Supervised Entity.

FirstExpiredCheckpointId_t	ID of the Checkpoint referenced in failed Supervision.
FirstExpiredAdditionalCheckpointId	ID of the Additional Checkpoint referenced in failed Supervision.
FirstExpiredModeId	ID of the WdgM Mode in which first Supervised entity got expired.
FirstElapsedCounterValue	Elapsed time (in ticks) in case of Deadline Supervision failure.
MainFunctID	ID of the WdgM Mainfunction to which first Supervised entity that got expired is mapped to.

Structure Element	No Failure	Alive Supervision Failure	Deadline Supervision Failure
FirstExpiredLocation_t	0	1	2 or 3
			2: Stop Checkpoint is not reached even though Maximum deadline duration since reporting of Start checkpoint has elapsed.
			3: Duration between Start and stop checkpoint reporting is outside the configured limits.
FirstExpiredSupervisedEntityId_t	Not applicable (Value will be '255' in this case).	X- ID of the first expired Supervised Entity.	X- ID of the first expired Supervised Entity.
FirstExpiredCheckpointId_t	Not applicable (Value will be '255' in this case).	Y - ID of the Checkpoint referenced in failed Alive Supervision.	Y - ID of the Checkpoint referenced as Start Checkpoint in failed Deadline Supervision.
FirstExpiredAdditionalCheckpointId	Not applicable (Value will be '255' in this case).	Not applicable (Value will be '255' in this case).	Z- ID of the Checkpoint referenced as Stop Checkpoint in failed Deadline Supervision.
FirstExpiredModeId	Not applicable (Value will be '255' in this case)	P - ID of the WdgM Mode in which first Supervised entity got expired.	P - ID of the WdgM Mode in which first Supervised entity got expired.

FirstElapsedCounterValue	Not applicable (Value will be '0' in this case).	Not applicable (Value will be '0' in this case).	Case FirstExpired-Location_t = 2:
			T - Duration (in ticks) since Start checkpoint was reported (measured at Api WdgM_MainFunction).
			Case FirstExpired-Location_t = 3:
			T- Duration (in ticks) between Start and Stop checkpoint reporting.
FirstExpiredAdditionalSupervisedEntityId	Not applicable (Value will be '255' in this case).	Not applicable (Value will be '255' in this case).	Not applicable (Value will be '255' in this case).
MainFunctID	Not applicable (Value will be '255' in this case)	M - ID of the WdgM Mode in which first Supervised entity got expired.	M - ID of the WdgM Mode in which first Supervised entity got expired.

4.8.8 WdgMRbRteDisabled

The WdgMRbRteDisabled configuration parameter can be used to control the Mode switch indication to RTE.

- If WdgMRbRteDisabled is set to TRUE, Mode switch indication to RTE is disabled.
- If WdgMRbRteDisabled is set to FALSE, Mode switch indication to RTE is left enabled and some additional RTE configuration is then required (see below).

4.8.8.1 RTE Configuration

If WdgMRbRteDisabled is set to FALSE (leaving Mode switch indication to RTE enabled), the Interfaces and Ports provided by WdgM are defined in the WdgM Software Component Description file WdgM_Cfg_SWCD.arxml. This file is auto-generated based on the configuration of WdgM.

Note that receiver port configurations are also required in the project to make use of WdgM interfaces via RTE.

4.8.9 WdgM_Rb_Cfg_IndicateCheckPointReached

WdgM_Rb_Cfg_IndicateCheckPointReached can be used for reporting Checkpoint reached for Alive/Deadline supervision of OS tasks in cases where WdgM_CheckpointReached cannot be scheduled directly in OS.

The two template files below should be adapted as described, and then included in the software build. Instances of WdgM_Rb_Cfg_IndicateCheckPointReached can then be scheduled at the appropriate location in OS.

4.8.9.1 Changes to “WdgM_Bsw_User.c”

In the name of the template function, replace “” with a suitable name (max 60 chars) to represent the Checkpoint to be reported.

In the second argument of the function call WdgM_CheckpointReached (WdgMConf_WdgMCheckpoint) replace “” with the Short name of the Check Point that is to be reported (max 60 chars).

In the first argument of the function call WdgM_CheckpointReached (WdgMConf_WdgMSupervisedEntity) replace “” with the Short name of the Supervised Entity to which the checkpoint belongs. Ensure that the complete name of the argument does not exceed 60 characters.

If more than one checkpoint is to be reported, create multiple instances of the template using the above steps for each instance.

4.8.9.2 Changes to “WdgM_Bsw_User.h”

This file contains the prototype of function WdgM_Rb_Cfg_IndicateCheckPointReached.

Update the name of the function to match the corresponding function name in “WdgM_Bsw_User.c”.

If more than one checkpoint is to be reported, create multiple instances of the function prototype.

Note: If WdgM_Rb_Cfg_IndicateCheckPointReached is not used or required in the project, either do not create the WdgM_Bsw_User.h and WdgM_Bsw_User.c files, or do not include them in the software build.

4.8.10 Configuring for Dem

If reporting of errors to DEM is enabled (WdgMDemStoppedSupervisionReport set to True), the reference to the following two DEM events must be configured -

- Supervision failed detect (WDGM_E_SUPERVISION)
- Mode switch failed detect (WDGM_E_SET_MODE)

The short name for the DEM events must be configured as WDGM_E_SUPERVISION and WDGM_E_SET_MODE to allow the Wdgm forward configuration to the correct DEM events.

4.8.11 Additional Monitoring Templates

The following macros are provided in the template file WdgM_Cfg_Additional-Monitoring.h_tpl, and are for the purpose of additional customised monitoring for WdgM_MainFunction by External Watchdog.

These also provide redundant recovery path in case of any possible operational failures of Internal Watchdogs, the additional customised monitoring of WdgM_-MainFunction by External Watchdog. The preprocessor switch WDGM_RB_ADDI-TIONAL_MONITORING_ENABLE shall be disabled in case the feature is not required.

These macros are to be called in the Master MainFunction context:

- WdgM_Rb_AdditionalMonitoringStart: To be mapped to appropriate customised function to start the monitoring.
- WdgM_Rb_AdditionalMonitoringStop:To be mapped to appropriate customised function for completion of the monitoring.

These macros are to be called in the Satellite MainFunction context:

- WdgM_Rb_AdditionalMonitoringStart_Satellite_(WdgMMainFunction_ShortName_Suffix):To be mapped to appropriate customised function to start the monitoring.
- WdgM_Rb_AdditionalMonitoringStop_Satellite_(WdgMMainFunction_ShortName_Suffix):To be mapped to appropriate customised function for completion of the moni-toring.

4.8.12 Error Handler Templates

There are two ways to include WdgM_Cfg_ErrorHandler_Callout.h and WdgM_Cfg_ErrorHandler_Callout.c in a project:

WdgM_Cfg_ErrorHandler_Callout.h_tpl and WdgM_Cfg_ErrorHandler_Call-out.c_tpl can be used to add user specific error reactions in case of Master alive supervision failure by Satellite Plus Mainfunction.

WdgM_Rb_ErrorHandler_Callout_SatPlus_*SatellitePlusPartition*() needs to be mapped to appropriate customised functions to enable the feature.

Generation of WdgM_Cfg_ErrorHandler_Callout.h_tpl and WdgM_Cfg_Er-rorHandler_Callout.c_tpl by a script in the out folder, which can then be placed at an appropriate path in the project for use (after removing _tpl).

Copying WdgM_Cfg_ErrorHandler_Callout.h_tpl and WdgM_Cfg_ErrorHan-dler_Callout.c_tpl from templates folder to the appropriate path in the project for use (after removing _tpl).

WdgM_Cfg_ErrorHandler_Callout.h_tpl and WdgM_Cfg_ErrorHandler_Call-out.c_tpl can be used to add user specific error reactions in case of Master alive supervision failure by Satellite Plus Mainfunction.

Replace the (MainFunction_short_name_suffix) portions with partition name of satellite plus or light master mainfunction.

Eg: Incase of Satellite plus Mainfunction (WdgMMainFunction_PartitionB) modify the callout as WdgM_Rb_ErrorHandler_Callout_SatPlus_PartitionB

For the memory sections enclosing the function definition and declaration, replace (Partition_Name) with the configured short name of corresponding partition

The callouts in the tpl files shall be modified by user for error handling by WdgM satellite plus or light master mainfunction.

NOTE: The error handling is only applicable when more than one MainFunction is configured. When only a single MainFunction is configured, then this will be Master only. The error handling feature is to be used with the Satellite function.

4.8.13 Memory Barrier Template

WdgM_Cfg_MemBarrier.h_tpl can be used to add user specific memory barrier based on compiler or hardware specific(assembly instruction) dependency.

The callout memory_barrier() shall be filled with compiler barrier instructions. Sample code is as below:

```
#define WDGM_SET_SEQUENCE_POINT()  __asm__ volatile ("" : : : "memory")
#define compiler_barrier()        WDGM_SET_SEQUENCE_POINT()
#define memory_barrier()         compiler_barrier()
```

memory_barrier() callout is treated as a “compiler_barrier” here because as per WdgM design one FIFO is allocated per task, as a result no cross-core/cross-partition write of FIFO is allowed.

4.8.14 Configuring Multiple MainFunctions

1. When using multiple mainfunctions configure **Partition Reference** and **WdgM Main Function Type** in the General Container.

In the general container configuration for Multiple MainFunctions configure the Master Mainfunction by setting **WdgMRbMainFunctionType** as **Master**:

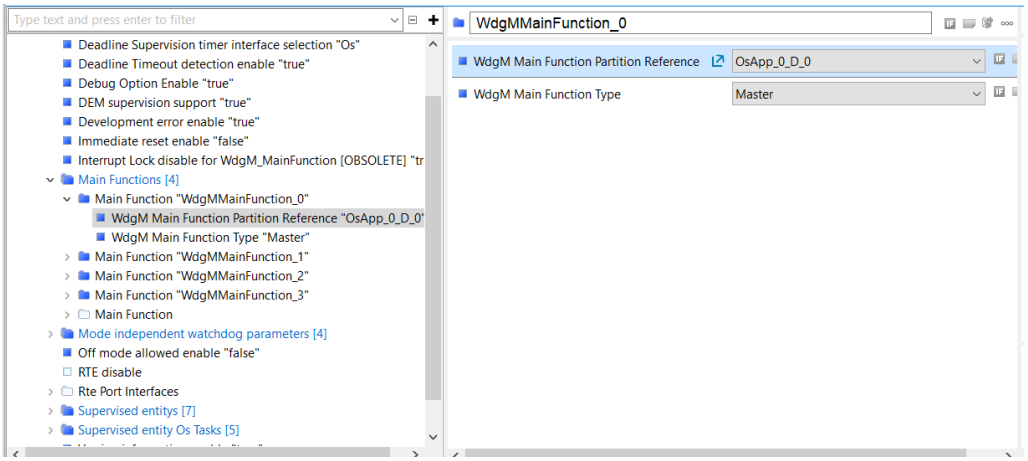

Figure 4.3.General Container Configuration for Multiple MainFunctions.

In the general container configuration for Satellite and Satellite Plus main functions. Configure the MainFunctions by setting Type **WdgMRbMainFunctionType** as **Satellite plus** (only one enhanced satellite) and remaining satellites as **Satellite**:

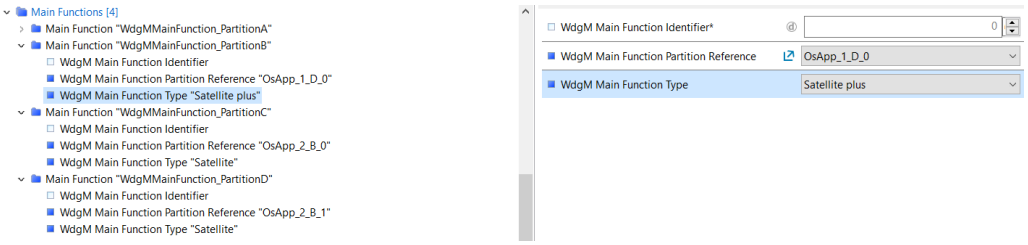

Figure 4.4.General Container Configuration for Satellite and Satellite Plus MainFunctions

2. Configure **Main Function Time Period** in **WdgM Mainfunction Mode Properties** under each WdgM Mode Container for multiple mainfunctions:

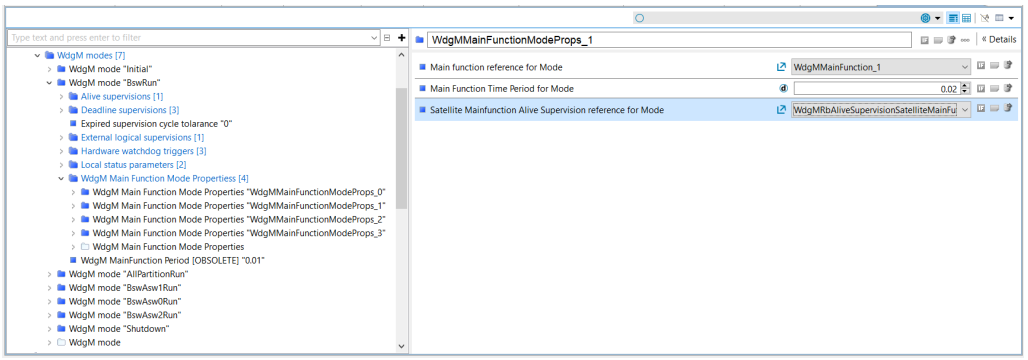

Figure 4.5.Configuration for Multiple MainFunctions in WdgMMode

3. Configure **Alive Supervision for Satellite MainFunctions** under each **WdgMConfigSet** container for each satellite mainfunction:

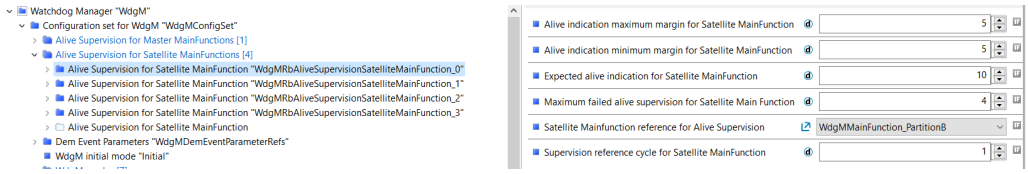

Figure 4.6.Configuration for Satellite Alive Supervision under WdgMConfigSet

> **NOTE**
>
> A warning is triggered if `WdgMRbSatelliteMainFunctionMinMargin` equals the `WdgMRbSatelliteMainFunctionExpectedAliveIndication`. It is generally recommended to set the `WdgMRbSatelliteMainFunctionMinMargin` lower than the `WdgMRbSatelliteMainFunctionExpectedAliveIndication` `.WdgMSatelliteSupervisionConfiguration` for Satellite alive supervision under `WdgMConfigSet`.

4. Configure **Alive Supervision for Master MainFunctions** under each **WdgM-ConfigSet** Container for Master mainfunction:

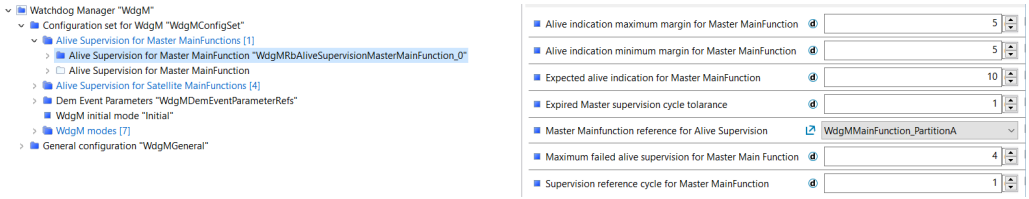

Figure 4.7.Configuration for Master Alive supervision under WdgMConfigSet

> **NOTE**
>
> A warning is triggered if the `WdgMRbMasterMainFunctionMinMargin` equals the `WdgMRbMasterMainFunctionExpectedAliveIndication`. It is generally recommended to set the `WdgMRbMasterMainFunctionMinMargin` lower than the `WdgMRbMasterMainFunctionExpectedAliveIndication` `.WdgMMasterSupervisionConfiguration` for Master alive supervision under `WdgMConfigSet`.

4.8.15 Configure Supervised Entity Os Tasks

1. Configure **Supervised entity Os Task** (WdgMRbOsTask) under the parent container.

2. Depends on the ASIL type selection:

Supervised Entity Os Tasks Configurations Multi-Asil

If ASIL type selection (`WdgMRbASILType`) is *WDGM_RB_MULTI_ASIL*, configure **Supervised entity Os Task reference** (`WdgMRbOsTaskRef`) with a reference to the corresponding Os Task in the OS Module for each **Supervised entity Os Tasks** (`WdgMRbOsTask`).

> **NOTE**
>
> The Os configuration parameter `OsTaskPeriod` should be configured for the corresponding Os Task in the AUTOSAR OS module (for fifo size calculations). **Hint:** In case of tasks where the task period is not applicable (e.g. Init, background tasks), it is recommended to configure the parameter **Maximum number of Checkpoints within one cycle of main function** (`WdgMRbOsTaskMaxCPReportPerMainFuncCycle`) considering the checkpoints and a sufficient buffer.

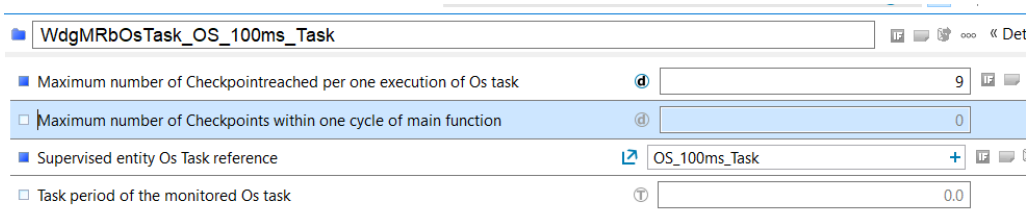

Figure 4.8.Supervised Entity Os Tasks Configurations Multi-Asil

Supervised Entity Os Tasks Configurations Same Asil

If the ASIL type selection (`WdgMRbASILType`) is *WDGM_RB_SAME_ASIL*, directly configure **Task period of the monitored Os Task** (`WdgMRbOsTaskPeriod`). **Hint:** In case of tasks where the task period is not applicable (e.g. Init, background tasks), it is recommended to configure the parameter **Maximum number of Checkpoints within one cycle of main function** (`WdgMRbOsTaskMaxCPReportPerMainFuncCycle`) considering the checkpoints and a sufficient buffer.

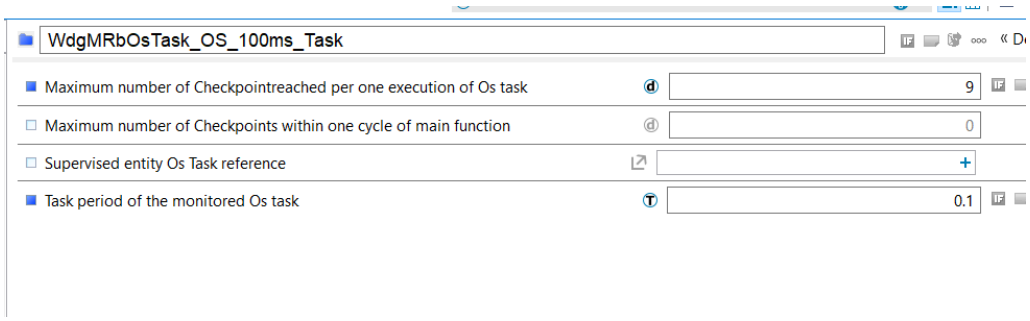

Figure 4.9.Supervised Entity Os Tasks Configurations Same Asil

3. Configure **Maximum number of Checkpoints within one cycle of main function** (WdgMRbOsTaskMaxCPReportPerMainFuncCycle) or **Maximum number of Checkpointreached per one execution of Os task** (WdgMRbOsTaskMaxCPReportPerOsTaskCycle), for each **Supervised entity Os Tasks** (WdgMRbOsTask). Refer to FIFO size calculations for more details on the configuration of these parameters.

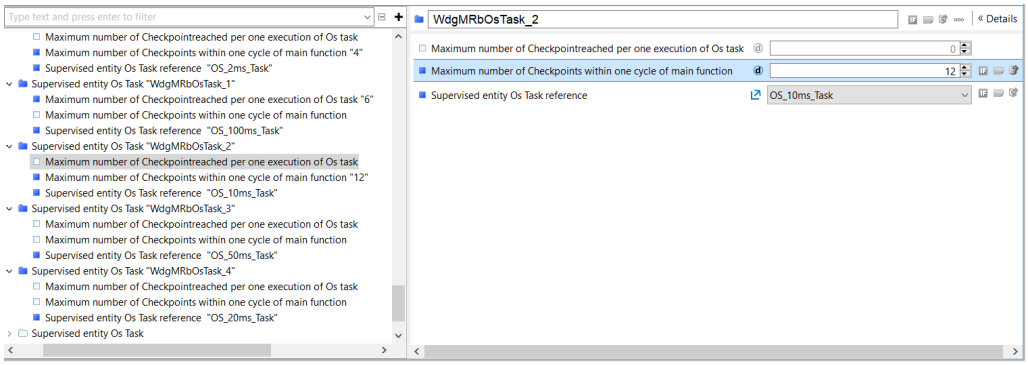

Figure 4.10.Supervised Entity Os Tasks Configurations

4.8.16 Mappings Related to Supervised Entity

1. Configure **Supervised entity** to MainFunction mapping (WdgM main function reference) and **Supervised entity** to Os task mapping (Supervised entity OsTask reference) in **Supervised entitys** container:

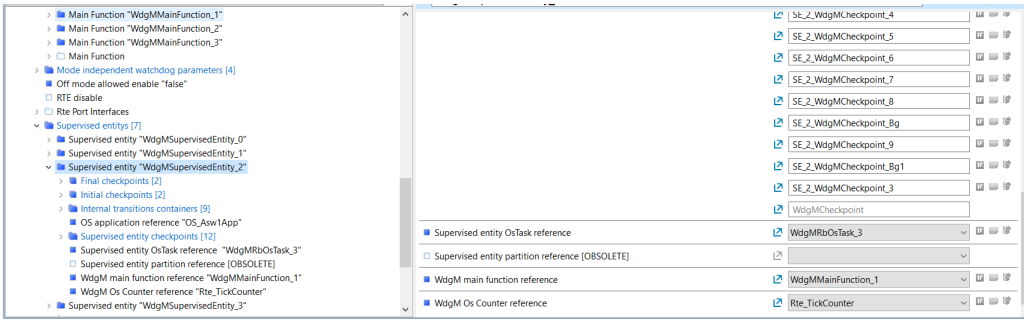

Figure 4.11.Mappings Related to Supervised Entity

4.8.17 WdgM FIFO Size Calculations

- Configuration of parameters **Maximum number of Checkpointreached per one execution of Os task** or **Maximum number of Checkpoints within one cycle of main function** should be taken care of when integrating. The configured value will be used by WdgM to calculate FIFO size for each configured Os tasks.
- If the parameters **Maximum number of Checkpointreached per one execution of Os task** or **Maximum number of Checkpoints within one cycle of main function** are not configured, then the default value shall be determined by WdgM for FIFO size for each configured Os tasks.
- For the configuration of parameter **Maximum number of Checkpoints within one cycle of main function** (WdgMRbOsTaskMaxCPReportPerMainFuncCycle), in case of the OsTask executing at a slower period than the WdgM MainFunction, it's recommended to add sufficient buffer to the value equal to Maximum number of checkpoints that will be reported in one cycle of the WdgM Mainfunction.
- For the configuration of parameter **Maximum number of Checkpoints within one cycle of main function** (`WdgMRbOsTaskMaxCPReportPerMainFuncCycle`), in the case of an OsTask executing at a faster period than the WdgM MainFunction, it's recommended to add a sufficient buffer in comparison to ((Maximum number of checkpoints that will be reported in one cycle of the monitored OsTask) x (Allowed maximum number of cycles of the monitored OsTask within one cycle of the WdgM MainFunction)). At least one additional cycle of the OsTask should be considered to allow the Maximum number of cycles of the monitored OsTask within one cycle of WdgM MainFunction.
- Consider checkpoints reported by all Supervised Entities corresponding to the particular OsTask.

Example 1

The configuration for parameter **Maximum number of Checkpoints within one cycle of main function** in Supervised Entity Os task container, In case of the OsTask (Time period : 10.0) executing faster than the WdgM MainFunction (Main-

functionperiod : 30.0) and total number of checkpoints of SEs mapped to same Os task is 2.

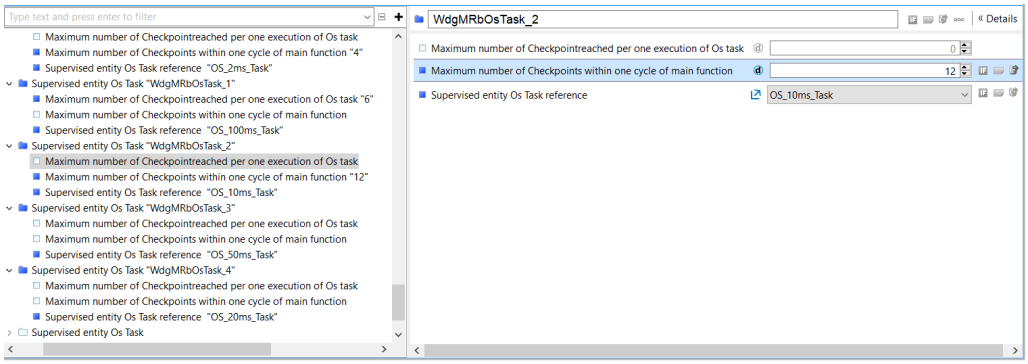

Figure 4.12.FIFO Size Related Configuration 1

Example 2

The configuration for parameter **Maximum number of Checkpoints within one cycle of main function** in Supervised Entity Os task container, In case of the OsTask(Time period : 100.0) executing slower than the WdgM MainFunction (Mainfunction period : 30.0) and total number of checkpoints of SEs mapped to same Os task is 9.

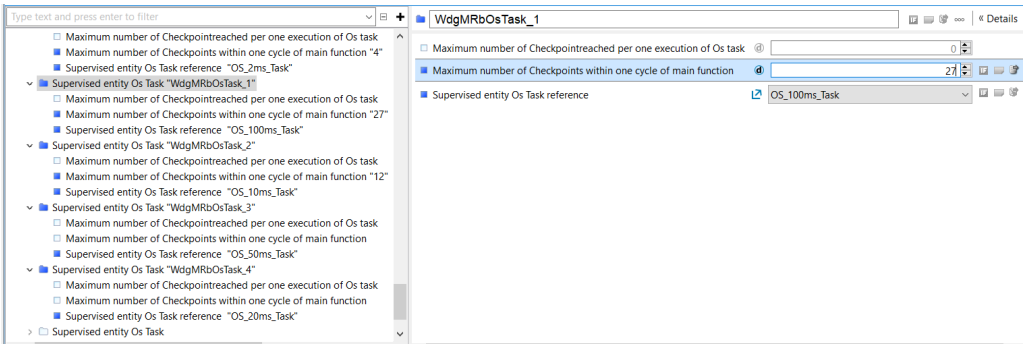

Figure 4.13.FIFO Size Related Configuration 2

4.9 Integration Advice

WdgM supports the generation of the WdgM_Cfg_BSWMD.arxml file. This file can be fed to ISOLAR-AB to generate the RTE header files.

4.9.1 Init Functions

The initialization of WdgM is performed by the EcuM module in all scenarios.

In the case of multi-ASIL or multi-partitions, partition-specific initialization is performed from the OS application context.

WdgM initialization must be performed after the initialization of the Diagnostic Event Manager (DEM) and the Development Error Tracer (DET).

WdgM does not initialize Wdg, which must be initialised before WdgM.

WdgM can only perform monitoring when the OS is active, so monitoring and Wdg triggering should not be configured to occur until the OS has been fully initialized.

Initialization of WdgM corresponding to Master partition should be done before the initialization of WdgM corresponding to Satellite partitions.

Init corresponding to Master partition handles the updating of the Global Supervision Status, Wdg Handling and setting Initial Mode.

Initialization of Supervision related Global variables and New Local Supervision Status are done in each partition specific Init.

The initialisation status of WdgM can be checked via the variable WdgM_GlobalStatus.

4.9.2 DeInit Functions

WdgM_DeInit and WdgM_PerformReset are used to deactivate WdgM.

4.9.2.1 WdgM_DeInit

WdgM is deactivated by WdgM_DeInit, which changes the WdgM Global Supervision Status to DEACTIVATED (Value = 4). This is done when WdgM is to be disabled.

- WdgM_DeInit should not be pre-empted by WdgM_SetMode or WdgM_MainFunction, or vice versa.
- WdgM_DeInit should only be called when the OS is no longer running.
- WdgM_MainFunction and WdgM_SetMode should not be called after WdgM_DeInit has been called.
- WdgM_DeInit should be called from the Master partition only when used in a safety related context.

4.9.2.2 WdgM_PerformReset

WdgM_PerformReset changes the WdgM Global Supervision Status to DEACTIVATED (Value = 4) and all internal Watchdog drivers are serviced with trigger value zero. This is done when the Controller is to be reset.

- WdgM_PerformReset should not be pre-empted by WdgM_SetMode or WdgM_MainFunction, or vice versa.
- WdgM_PerformReset, WdgM_SetMode and WdgM_MainFunction should be called sequentially in the same OS task.
- WdgM_MainFunction and WdgM_SetMode should not be called after WdgM_PerformReset has been called.
- WdgM_PerformReset should be called from the Master partition only when used in a safety related context.

4.9.3 Scheduled Functions

WdgM_Init

- `WdgM_Init` must be initialised/called by EcuM via the driver configuration. This is should be done for all scenarios (i.e. multi-partition or no partition).
    - *Single/no-partition:* In the case of a single or no partition scenario, initialization is via EcuM.
    - *Multi-partition:* In the case of multi-partition, `WdgM_Init(const WdgM_ConfigType * ConfigPtr)` will be used for variant handling. It should be called by the EcuM configuration. For each partition (Master and Satellites), one Init event will be configured.
- Every Init event should be mapped to the Runnable entity of the corresponding Init function `WdgM_Init_<PartitionName>`.
- Each Init Event should be configured under the OS *ini* task of its respective partition through RTE. e.g. the Init Event "I_WdgM_Init_0B1" should be configured under "OS_0B1_Ini_Task" via RteEventToTaskMapping.
- To be taken care by integrator that initialization of WdgM corresponding to Master partition should be done before initialization of WdgM corresponding to Satellite Partitions.

WdgM_MainFunction

WdgM_MainFunction is called at a cyclic interval (defined by the WdgMSupervisionCycle parameter), and it is responsible for performing monitoring analysis and providing servicing information to Wdg via WdgIf.

- If RTE is Enabled: Based on the configurations on "Main Function Period" the RTE generates a task to call WdgM_MainFunction based on different timing requirements. Care should be taken to schedule the generated RTE API based on the lowest timing raster. For example, if the Main Function timing period is "10ms, 20ms, 100ms", then schedule the RTE-generated API in a 10ms task raster.
- If RTE is Disabled: Make sure all the "Run" modes have the same timing period and schedule WdgM_MainFunction in the task raster as configured.
- When scheduling Multiple MainFunctions then master mainfunction should be scheduled in master partition and satellite mainfunction scheduled in satellite partitions.
- If WdgM Modes have different WdgM Main function Periods (e.g. 10ms, 100ms etc) then RTE will call WdgM_MainFunction on different rasters based on the current mode of WdgM.
- WdgM_MainFunction can pre-empt WdgM_SetMode. During Mode Change, if the new mode has a schedule period that is greater than the previous mode Wdg Trigger value, then there is a possibility that Wdg will cause a reset. To avoid this, WdgM_MainFunction and WdgM_SetMode should be executed sequentially.
- WdgM can perform cyclic triggering of Wdg as long as the OS task (i.e.

scheduler) is active. If the scheduler is inactive, WdgM cannot perform Wdg triggering. In such cases it is recommended to place stub codes or API call-outs to trigger Wdg during shutdown.

- WdgM assumes that Wdg will take care of accessing cross core drivers.

WdgM_CheckpointReached

- This API performs reporting of status (i.e. the point in code to be monitored or Checkpoint) to WdgM. The WdgM uses this status information to evaluate system status every time WdgM_MainFunction is called
- Partition specific WdgM_CheckpointReached needs to be placed in appropriate partitions with respective arguments at the Checkpoint (i.e. locations in code) which are to be monitored.
- FIFO mechanism is used for Checkpoint reached information sharing with WdgM_MainFunction.

WdgM_SetMode

- The API is used to change the mode of WdgM to enable, disable or change the monitoring pre-requisites based on each mode.
- WdgM_SetMode() will be synchronous. New mode will be applied in a partition after first time execution of partition specificMainFunction since Mode change.
- The integrator shall call WdgM_SetMode API from Master partition only(this is the assumption of use in a safety related context).
- **Mode Change limitation:**
    - WdgM_SetMode will be callable in partition corresponding to WdgM_MainFunction in a sequential manner only.

WdgM_DeInit

The deinit changes WdgM Global Supervision Status to DEACTIVATED i.e. Value = 4. This is done when WdgM is to be disabled.

WdgM_DeInit API calling has limitations.

WdgM_DeInit should not be pre-empted by API WdgM_SetMode and WdgM_-MainFunction vice versa i.e WdgM_DeInit shall not pre-empt WdgM_SetMode and WdgM_MainFunction.

Recommended: WdgM_DeInit should be called when Operating System is not running.

After the call of WdgM_DeInit, WdgM_MainFunction and WdgM_SetMode should not be called.

The integrator shall call WdgM_Deinit API from Master partition only (this is the assumption of use in a safety related context).

WdgM_PerformReset

The PerformReset changes WdgM Global Supervision Status to DEACTIVATED i.e. Value = 4 and all internal Watchdog drivers are serviced with trigger value zero.

This is done when Controller is to be reset.

The integrator shall call WdgM_PerformReset API from Master partition only (this is the assumption of use in a safety related context).

WdgM_PerformReset API calling has limitations.

WdgM_PerformReset should not be pre-empted by API WdgM_SetMode and WdgM_MainFunction vice versa i.e WdgM_PerformReset shall not pre-empt WdgM_SetMode and WdgM_MainFunction.

Recommended: WdgM_PerformReset, WdgM_SetMode and WdgM_MainFunction should be called in same OS task (i.e WdgM_PerformReset, WdgM_Setmode and WdgM_MainFunction should be called sequentially).

After the call of WdgM_PerformReset, WdgM_MainFunction and WdgM_SetMode should not be called.

4.9.4 Multi-Partition Integration

When configuring multi-partition in WdgM, additional configuration steps are required within and outside of WdgM.

- Using partitioning provides a method to separate software components with different access rights (e.g., due to ASIL-levels) from each other and so to ensure freedom from interference. Within a partition the software components run in an isolated area which prevents unwanted interference of other partitions. Therefore all RAM structures required for the Master/Satellite interaction must be moved into the master's/satellite's partition RAM area (RAM Split). Refer to the Memory mapping for mapping of RAM variables.
- When integrating an MPU (Memory Protection Unit) this should be setup to check that the satellite-partition or master-partition unintended writes to RAM occur. Within WdgM it is assumed that either such a protection is in place, or there is no safety-relevant impact if data is overwritten.
- The WdgM Multi-Asil feature partitions the SWCD and BSWMD. As a result, the Master Partition will generate SWCD and BSWMD with the short name "WdgM", whereas the Satellite Partitions will be labeled as "WdgM_{PartitionName}". The mapping of the WdgM initialization must be managed through SWCD INIT-EVENTS, as the EcuM is unable to handle the initialization of BswEents that are specific to each partition.

Note: The RTE Option "–deviate-bsw-any-partition=1" must be enabled when using the WdgM Multi Partition feature, as the RTE does not allow BswEvents instances to be mapped to tasks that can execute in any partition.

4.9.5 Correct Pointer Passed to WdgM Get APIs

Passing incorrect pointers (pointers carrying an address belonging to an unintended partition) to the WdgM-interfaces is still a risk for FFI (Freedom From Interference). When a pointer of a incorrect partition is passed by the user, WdgM will ultimately try to write to this location, and as the system provides a RAM split the user should configure the MPU so this can be detected.

The Following are WdgM Get APIs (commonly represented as WdgM_GetXXX())

to which usually a pointer is passed to capture the WdgM data info.

- WdgM_GetGlobalStatus
- WdgM_GetLocalStatus
- WdgM_RB_GetMasterStatus
- WdgM_GetMode
- WdgM_GetFirstExpiredSEID
- WdgM_RB_GetFirstExpiredMainFunctID
- WdgM_GetVersionInfo

4.9.5.1 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The WdgM module does not use any exclusive OS resources.

4.10 Hardware Requirements

Table 4.83.WdgM Hardware Requirements

Support	Usage
FLASH	3280 bytes
RAM	75 bytes

Usage calculated based on a sample configuration.

5 Glossary

ASIC
Application Specific Integrated Circuit

ASW
Autosar Software

BswM
AUTOSAR BSW Mode Manager module

CP Checkpoint

CRC
Cyclic Redundancy Checks

Crc
The Crc module, part of the RTA-SAFE stack

E2E
AUTOSAR End-to-End Communication Protection module

E2EXf
AUTOSAR End-to-End Communication Protection Transformer module

RTE
AUTOSAR Run-Time Environment, implements the AUTOSAR Virtual Functional Bus interfaces and thereby realizes the communication between SWCs

SE Supervised Entity

SEID
Supervised Entity Identifier

SWC
Software Component

Wdg
Watchdog driver

WdgIf
AUTOSAR Watch Dog Interface module

WdgM
AUTOSAR Watch Dog Manager module